2.3 Implicity Defined Attributes

Contents

2.3.1a The Minimum of a Function

Overview

- We have defined an attribute $a(\mathcal{P})$ to be a summary of the population $\mathcal{P}$. The numeric and graphical attributes that we have discussed so far are called **explicit** attributes because they are defined *explicitly* by the population variates.
- In this section we will discuss **implicitly** defined attributes, which also summarize the population $\mathcal{P}$, but which are defined only implicitly by the population variates. In particular, such attributes are solutions to an optimization problem (i.e., the solution to an equation or a set of equations)
- We take an in-depth look at linear regression as a context for discussion of implicitly defined attributes and their properties.

In most practical situations we are interested in a (possibly vector-valued) attribute $\boldsymbol{\theta}$ which minimizes some function $\rho(\boldsymbol{\theta}; \mathcal{P})$ of the variates in the population.

- That is, we want the value $\widehat{\boldsymbol{\theta}}$ which satisfies

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}}{\operatorname{argmin}} \, \rho(\boldsymbol{\theta}; \mathcal{P})$$

where the possible values of $\boldsymbol{\theta}$ may be constrained to be in some set $\boldsymbol{\Theta}$.

- Note that maximizing a function is the same as minimizing its negation:

$$- \max_{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}} \rho(\boldsymbol{\theta}; \mathcal{P}) = \min_{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}} -\rho(\boldsymbol{\theta}; \mathcal{P})$$

and so

$$\underset{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}}{\operatorname{argmax}} \, \rho(\boldsymbol{\theta}; \mathcal{P}) = \underset{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}}{\operatorname{argmin}} -\rho(\boldsymbol{\theta}; \mathcal{P})$$

Therefore, we only need to consider minimization here.

- The most common form for $\rho(\boldsymbol{\theta}, \mathcal{P})$ is a sum of functions $\rho(\boldsymbol{\theta}, u)$ evaluated at each unit $u \in \mathcal{P}$:

$$\rho(\boldsymbol{\theta}, \mathcal{P}) = \sum_{u \in \mathcal{P}} \rho(\boldsymbol{\theta}, u)$$

The following examples include attributes defined implictly in this way. You will be familiar with them already, but you probably have not defined them in this way.

Examples (scalar valued attributes)

Some familiar examples for a **scalar valued** attribute $\theta \in \mathbb{R}$ and $u \in \mathcal{P}$ include:

- **Least-squares**: If $\rho(\theta; u) = (y_u - \theta)^2$ then

$$\begin{aligned}
\widehat{\theta} &= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}}\, \rho(\theta, \mathcal{P}) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} \rho(\theta, u) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} (y_u - \theta)^2 \\
&= \overline{y}
\end{aligned}$$

- **Weighted least-squares**: If $\rho(\theta; u) = w_u (y_u - \theta)^2$ then

$$\begin{aligned}
\widehat{\theta} &= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}}\, \rho(\theta, \mathcal{P}) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} \rho(\theta, u) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} w_u (y_u - \theta)^2 \\
&= \frac{\sum_{u \in \mathcal{P}} w_u y_u}{\sum_{u \in \mathcal{P}} w_u}
\end{aligned}$$

- **Least absolute deviations**: If $\rho(\theta; u) = |y_u - \theta|$ then

$$\begin{aligned}
\widehat{\theta} &= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}}\, \rho(\theta, \mathcal{P}) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} \rho(\theta, u) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} |y_u - \theta| \\
&= Q_y(1/2)
\end{aligned}$$

- **Least generalized-absolute deviations**: If for some $q \in (0, 1)$ we define the `vee` function

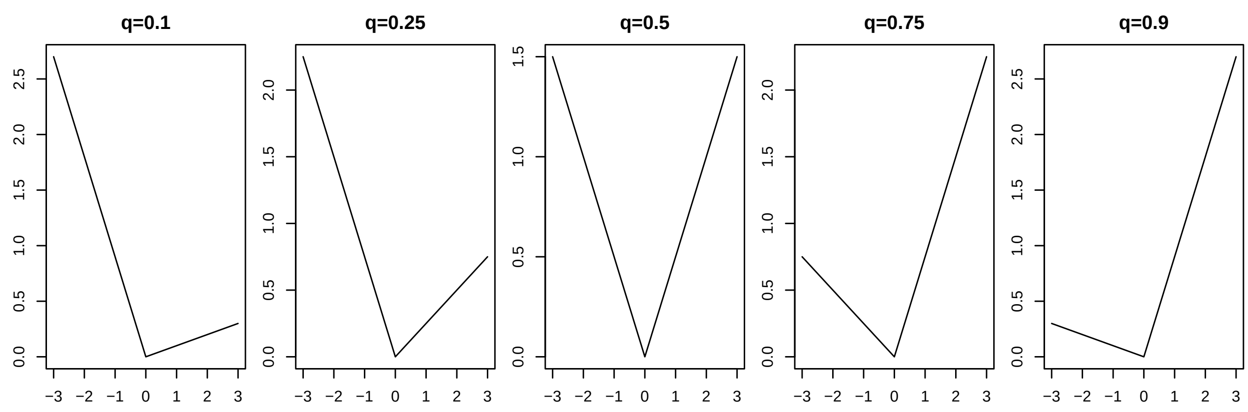
$$\rho_q(\theta; u) = \begin{cases} q(y_u - \theta) & \text{if } y_u \geq \theta \\ (q-1)(y_u - \theta) & \text{if } y_u < \theta \end{cases}$$

  then

$$
\begin{aligned}
\widehat{\theta} &= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}}\, \rho(\theta, \mathcal{P}) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} \rho(\theta, u) \\
&= \underset{\theta \in \mathbb{R}}{\operatorname{argmin}} \sum_{u \in \mathcal{P}} \rho_q(\theta; u) \\
&= Q_y(q)
\end{aligned}
$$

Here are several examples of the `vee` function for different values of q.

```
vee <- function(z, q) {
    val = q * z
    val[z < 0] = (q - 1) * z[z < 0]
    return(val)
}

z = seq(-3, 3, 0.01)
par(mfrow = c(1, 5), mar = 2.5 * c(1, 1, 1, 0.1))
plot(z, vee(z, q = 1/10), main = "q=0.1", type = "l")
plot(z, vee(z, q = 1/4), main = "q=0.25", type = "l")
plot(z, vee(z, q = 2/4), main = "q=0.5", type = "l")
plot(z, vee(z, q = 3/4), main = "q=0.75", type = "l")
plot(z, vee(z, q = 9/10), main = "q=0.9", type = "l")
```



Below we illustrate that the 30^{th} quantile, $Q_y(0.3)$, can be determined by minimizing a sum of appropriate `vee` functions. Note that in principle this method can be used to calculate *any* quantile.

1. Set the randomization seed using the commdand: `set.seed(341)`. Doing so ensures that all simulations give the same answer when you rerun them.

2. Simulate a population.

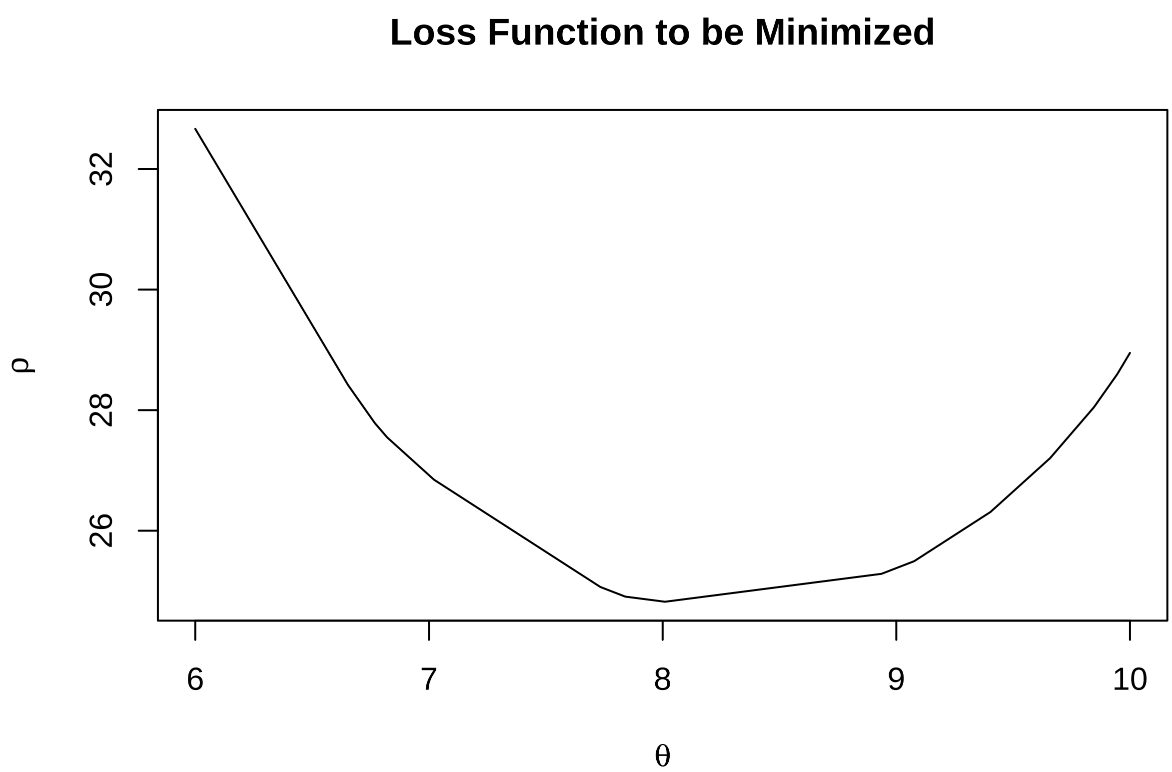
```r
set.seed(341)
y.pop <- rnorm(25, 10, 3)
```

3. Define the loss function (sum of `vee` functions) that is to be minimized.

```r
rho.fun <- function(theta, q.val) {
    return(sum(sapply(X = y.pop - theta, FUN = vee, q = q.val)))
}
```

4. Plot the function.

```r
theta = seq(6, 10, length.out = 1000)
rho.vals = numeric(length(theta))
for (i in 1:length(theta)) {
    rho.vals[i] = rho.fun(theta[i], q.val = 0.3)
}
plot(theta, rho.vals, type = "l", xlab = bquote(theta), ylab = bquote(rho[]),
    main = "Loss Function to be Minimized")
```



5. Use a numerical minimizer in `R` to find the argument that minimizes this function. We will use all three of `nlminb`, `optimize`, and `optim` for purposes of illustration.

```r
nlminb(start = 0.5, objective = rho.fun, q.val = 0.3)
```

```
## $par
## [1] 8.010171
##
## $objective
## [1] 24.82168
##
## $convergence
## [1] 0
##
## $iterations
## [1] 12
##
## $evaluations
## function gradient
##       22       14
##
## $message
## [1] "X-convergence (3)"
```

```r
optimize(f = rho.fun, interval = range(y.pop), q.val = 0.3)
```

```
## $minimum
## [1] 8.010159
##
## $objective
## [1] 24.82169
```

```r
optim(par = 0.5, fn = rho.fun, q.val = 0.3)
```

```
## $par
## [1] 8.010168
##
## $value
## [1] 24.82169
##
## $counts
## function gradient
##       52       NA
##
## $convergence
## [1] 0
##
## $message
## NULL
```

Compare these results to the built-in `quantile` function:

```r
quantile(y.pop, 0.3)
```

```
##      30%
## 8.195491
```

```r
quantile(y.pop, 0.3, type = 1)
```

```
##      30%
## 8.010171
```

Example (vector valued attributes): Simple Linear Regression

A familiar **vector valued** attribute is the vector of coefficients associated with the following simple linear regression:

$$y_u = \alpha + \beta\,(x_u - c) + r_u \qquad \forall\; u \in \mathcal{P}$$

The attribute of interest is $\boldsymbol{\theta} = (\alpha, \beta)$.

Note that a re-centering of the x_u values in a linear regression is not uncommon. Typically c is chosen to be a meaningful value in the data set such as the average x_u value (i.e., $c = \bar{x}$), for example. Different choices of c give rise to different interpreations for α. Not all such interpretations have practical relevance.

- These coefficients are determined implicitly by

$$\widehat{\boldsymbol{\theta}} = \left(\widehat{\alpha}, \widehat{\beta}\right) = \underset{(\alpha,\beta)\;\in\;\mathbb{R}^2}{\operatorname{argmin}} \sum_{u\;\in\;\mathcal{P}} \left(y_u - \alpha - \beta(x_u - c)\right)^2$$

- It can be shown (show this) that

$$\widehat{\alpha} = \bar{y} - \widehat{\beta}\;(\bar{x} - c) \qquad \text{and} \qquad \widehat{\beta} = \frac{\sum_{u\in\mathcal{P}}(x_u - \bar{x})(y_u - \bar{y})}{\sum_{u\in\mathcal{P}}(x_u - \bar{x})^2}$$

- The resulting estimates determine the **least-squares** fitted line:

$$y = \widehat{\alpha} + \widehat{\beta}\;(x - c)$$

- The equation of the fitted values, defined for all $u \in \mathcal{P}$, is:

$$\widehat{y}_u = \widehat{\alpha} + \widehat{\beta}\,(x_u - c)$$

- The residuals are

$$\widehat{r}_u = y_u - \widehat{\alpha} - \widehat{\beta}\,(x_u - c)$$

Each residual is the signed vertical distance between the point (x_u, y_u) and the point $(x_u, \widehat{y}_u) = \left(x_u, \widehat{\alpha} + \widehat{\beta}\,(x_u - c)\right)$. The latter point is the value of the fitted line, defined by $\widehat{\boldsymbol{\theta}} = \left(\widehat{\alpha}, \widehat{\beta}\right)$, calculated at $x = x_u$.

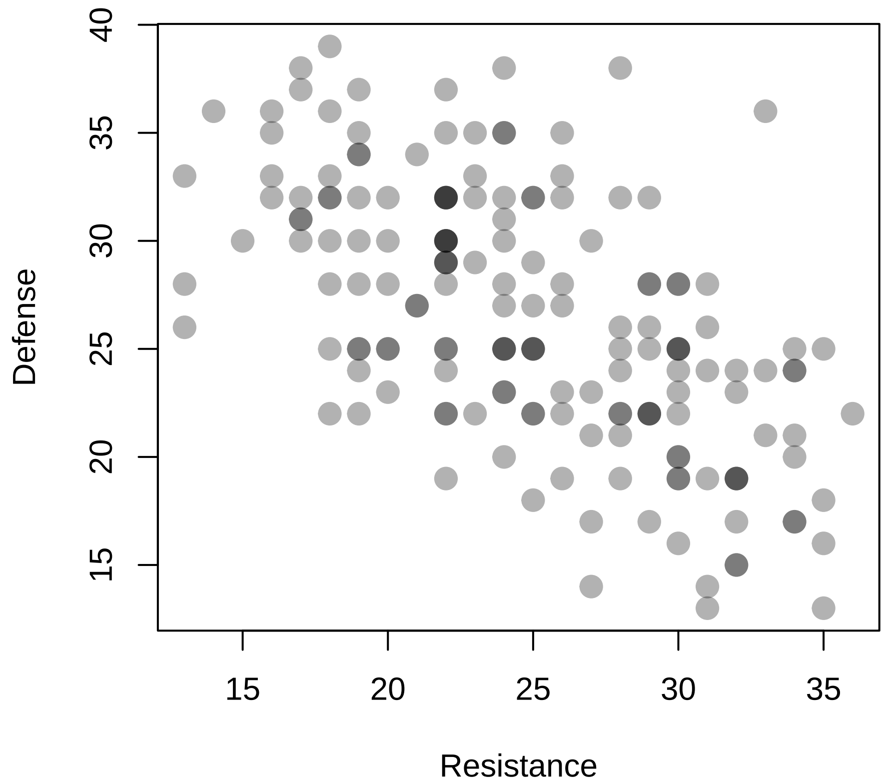
Fire Emblem Heroes Example (Calculating Influence)

- `RES`: resistance, ability to absorb magical attack (use as x)
- `DEF`: defense, ability to absorb physical attacks (use as y)

```
feh = read.csv("../../Data/feh.csv", header = TRUE)
head(feh)
```

```
##      Name       Type     Move HP ATK SPD DEF RES Total
## 1    Abel Blue Lance  Cavalry 39  33  32  25  25   154
## 2 Alfonse  Red Sword Infantry 43  35  25  32  22   157
## 3     Alm  Red Sword Infantry 45  33  30  28  22   158
## 4  Amelia  Green Axe  Armored 47  34  34  35  23   173
## 5    Anna  Green Axe Infantry 41  29  38  22  28   158
## 6  Arthur  Green Axe Infantry 43  32  29  30  24   158
```

```
plot(feh$RES, feh$DEF, main = "", pch = 19, cex = 1.5, col = adjustcolor("black",
    alpha = 0.3), xlab = "Resistance", ylab = "Defense")
```



- The averages for defense and resistance, respectively, are $\bar{y} = 26.4$ and $\bar{x} = 24.9$
- This relationship can be summarized as

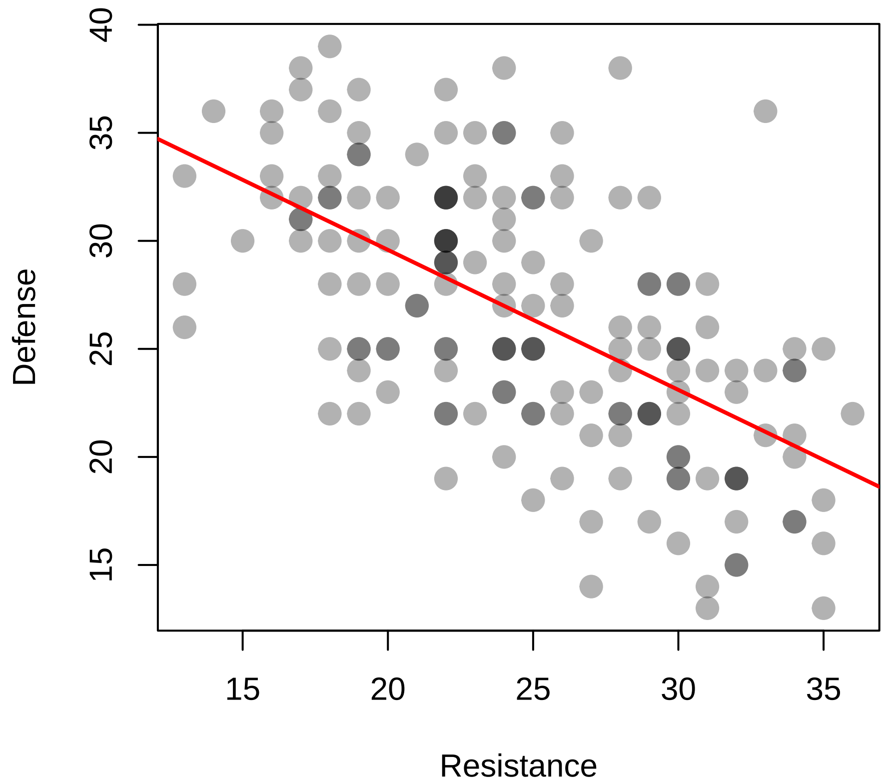
$$\text{Defense} = 42.5 - 0.65 \times \text{Resistance} + \text{error}$$

  - or equivalently as

$$\text{Defense} = 26.4 - 0.65 \times [\text{Resistance} - 24.9] + \text{error}$$

  - both equations (non-centred and centered, respectively) yield the same line and fitted values.

```
par(mfrow = c(1, 1))

plot(feh$RES, feh$DEF, main = "", pch = 19, cex = 1.5, col = adjustcolor("black",
    alpha = 0.3), xlab = "Resistance", ylab = "Defense")

fit = lm(DEF ~ RES, data = feh)
abline(fit, col = "red", lwd = 2)
```



- What have we done here?

- Does it make sense to use a regression line here?

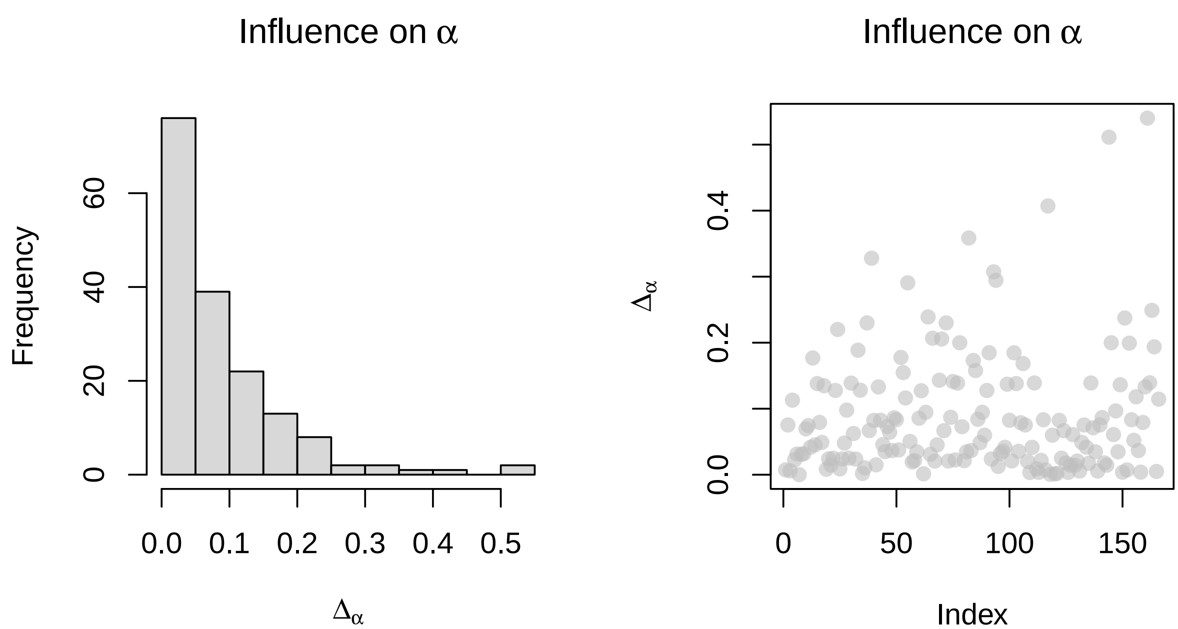
- In terms of the game is the regression line important?

- Is the notion of **influence** still relevant with implicitly defined attributes?

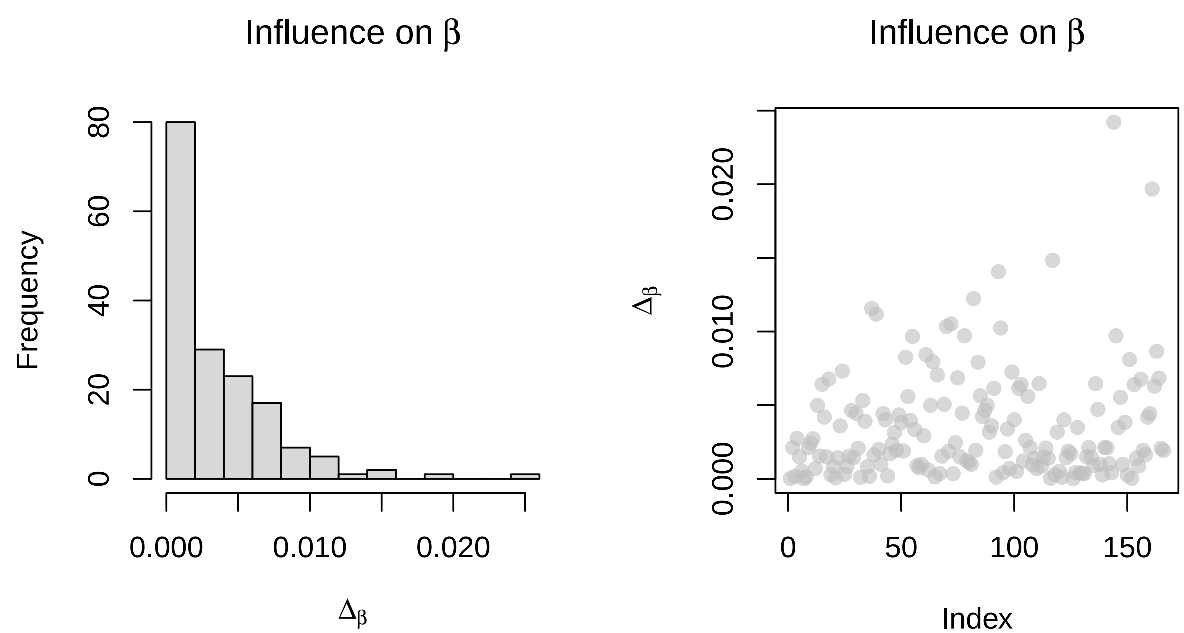
```r
N = nrow(feh)
delta = matrix(0, nrow = N, ncol = 2)
for (i in 1:N) {
    ## feh[-i] removes the ith row from a vector
    fit.no.i = lm(DEF ~ RES, data = feh[-i, ])
    delta[i, ] = abs(fit$coef - fit.no.i$coef)
}
```

Then plot the Δ's associated with each of the parameters (i.e., each component of the attribute $\boldsymbol{\theta}$)

```r
par(mfrow = c(1, 2))
hist(delta[, 1], breaks = "FD", main = bquote("Influence on" ~ alpha), xlab = bquote(Delta[alpha]),
    col = adjustcolor("grey", 0.6))
plot(delta[, 1], ylab = bquote(Delta[alpha]), main = bquote("Influence on" ~
    alpha), pch = 19, col = adjustcolor("grey", 0.6))
```

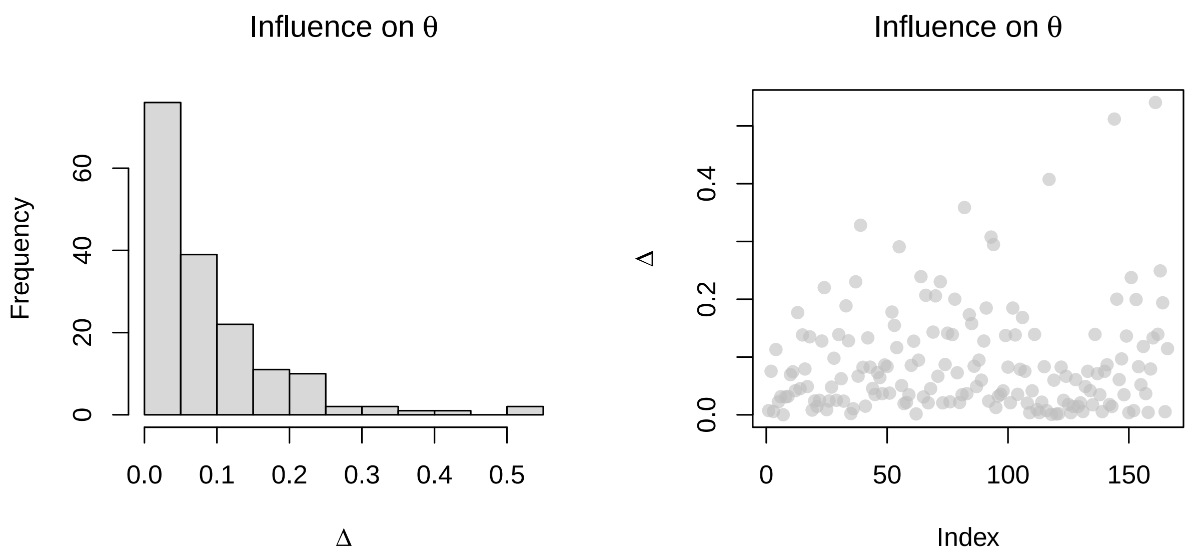



```r
par(mfrow = c(1, 2))
hist(delta[, 2], breaks = "FD", main = bquote("Influence on" ~ beta), xlab = bquote(Delta[beta]),
    col = adjustcolor("grey", 0.6))
plot(delta[, 2], ylab = bquote(Delta[beta]), main = bquote("Influence on" ~
    beta), pch = 19, col = adjustcolor("grey", 0.6))
```



These plots illustrate the inluence each unit has on the indvidual coefficient estimates, but they don't illustrate the influence each unit has on the regression line as a whole. To do this we define influence for a vector valued attribute using the vector's Euclidean norm.

```
par(mfrow = c(1, 2))

delta2 = apply(X = delta, MARGIN = 1, FUN = function(z) {
    sqrt(sum(z^2))
})
hist(delta2, breaks = "FD", main = bquote("Influence on" ~ theta), xlab = bquote(Delta),
    col = adjustcolor("grey", 0.6))
plot(delta2, main = bquote("Influence on" ~ theta), ylab = bquote(Delta), pch = 19,
    col = adjustcolor("grey", 0.6))
```



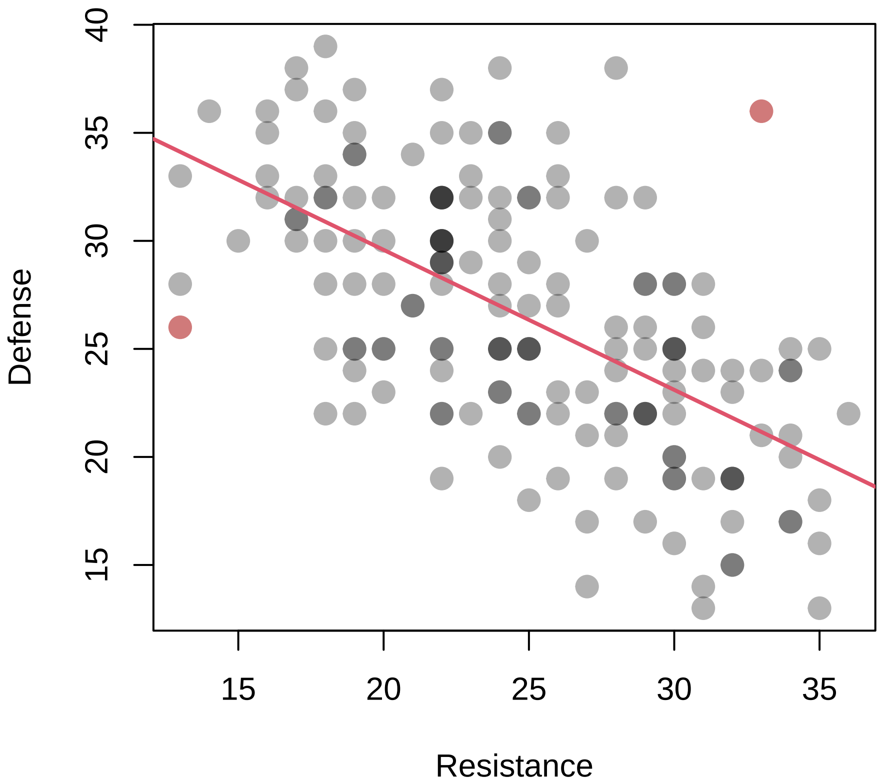
The two units with the largest influence are:

```
feh[delta2 > 0.5, ]
```

```
##       Name        Type     Move HP ATK SPD DEF RES Total
## 144 Sheena   Green Axe  Armored 45  30  25  36  33   169
## 161 Virion Neutral Bow Infantry 46  31  31  26  13   147
```

These are the weakest and one of the stronger characters in terms of resistance. Let's highlight them on the scatterplot.

```
col.nam = c(adjustcolor("black", alpha = 0.3), adjustcolor("firebrick", alpha = 0.6))

plot(feh$RES, feh$DEF, main = "", pch = 19, cex = 1.5, col = col.nam[(delta2 >
    0.5) + 1], xlab = "Resistance", ylab = "Defense")

# fit = lm(DEF ~ RES, data=feh)
abline(fit, col = 2, lwd = 2)
```



- How else could we define influence for vector valued attributes?

Summary

- We defined an attribute, $\boldsymbol{\theta}$, as the value which minimizes some function $\rho(\boldsymbol{\theta}; \mathcal{P})$ of the variates in the population.

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}}{\operatorname{argmin}} \, \rho(\boldsymbol{\theta}; \mathcal{P})$$

- We examined this type of attribute in the context of simple linear regression. We also reviewed the concept of influence within this new paradigm.

Exercises

From the Implicit Attribute exercises try problem 2.

2.3.1b Dealing with Influential Units in Linear Regression

Overview

- Here we will illustrate a number of methods for dealing with influential units in the context of simple linear regression.
  - We do so in the context of a dataset of body and brain weights from a variety of animals.
- In particular, we consider removing the influential unit(s) and alternative forms of regression, including:
  - weighted least squares and robust regression, both of which provide alternative ways of characterizing the relationship between body and brain weight.
- We also use this dataset as an opportunity to revisit the usefulness of power transformations.

In this section we discuss the identification of units that have a large influence on regression coefficients and illustrate several ways to mediate this influence. We consider deleting them, and also how to design attributes (i.e., alternative methods of estimating regression coefficients) that are resistant to the effect of influential observations.

Removing Influential Units

Animals Example

The data we consider here is the `Animals2` dataset from the `robustbase` package in `R`. The data frame contains average brain and body weights for $N = 65$ species of land animals:

- `body` represents body weight in `kg`
- `brain` represents brain weight in `grams`

An excerpt of the data as well as numerical and graphical summaries are shown below:

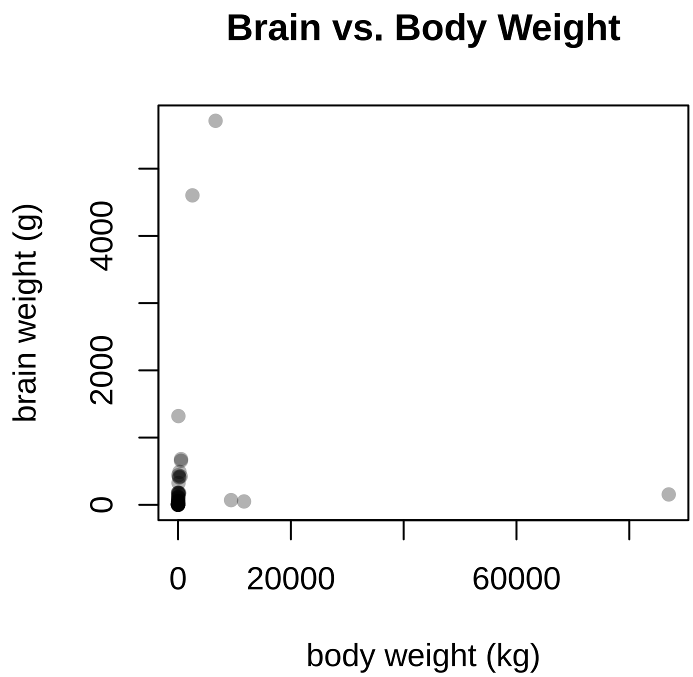
```
library(robustbase)
head(Animals2, n = 5)
```

```
##                    body brain
## Mountain beaver   1.35   8.1
## Cow             465.00 423.0
## Grey wolf        36.33 119.5
## Goat             27.66 115.0
## Guinea pig        1.04   5.5
```

```
summary(Animals2)
```

```
##       body               brain
##  Min.   :    0.00   Min.   :   0.14
##  1st Qu.:    0.75   1st Qu.:   5.00
##  Median :    3.50   Median :  21.00
##  Mean   : 1852.69   Mean   : 274.29
##  3rd Qu.:   60.00   3rd Qu.: 157.00
##  Max.   :87000.00   Max.   :5712.00
```

```
plot(Animals2, main = "Brain vs. Body Weight", xlab = "body weight (kg)", ylab = "brain weight (g)",
    pch = 16, col = adjustcolor("black", alpha.f = 0.3))
```



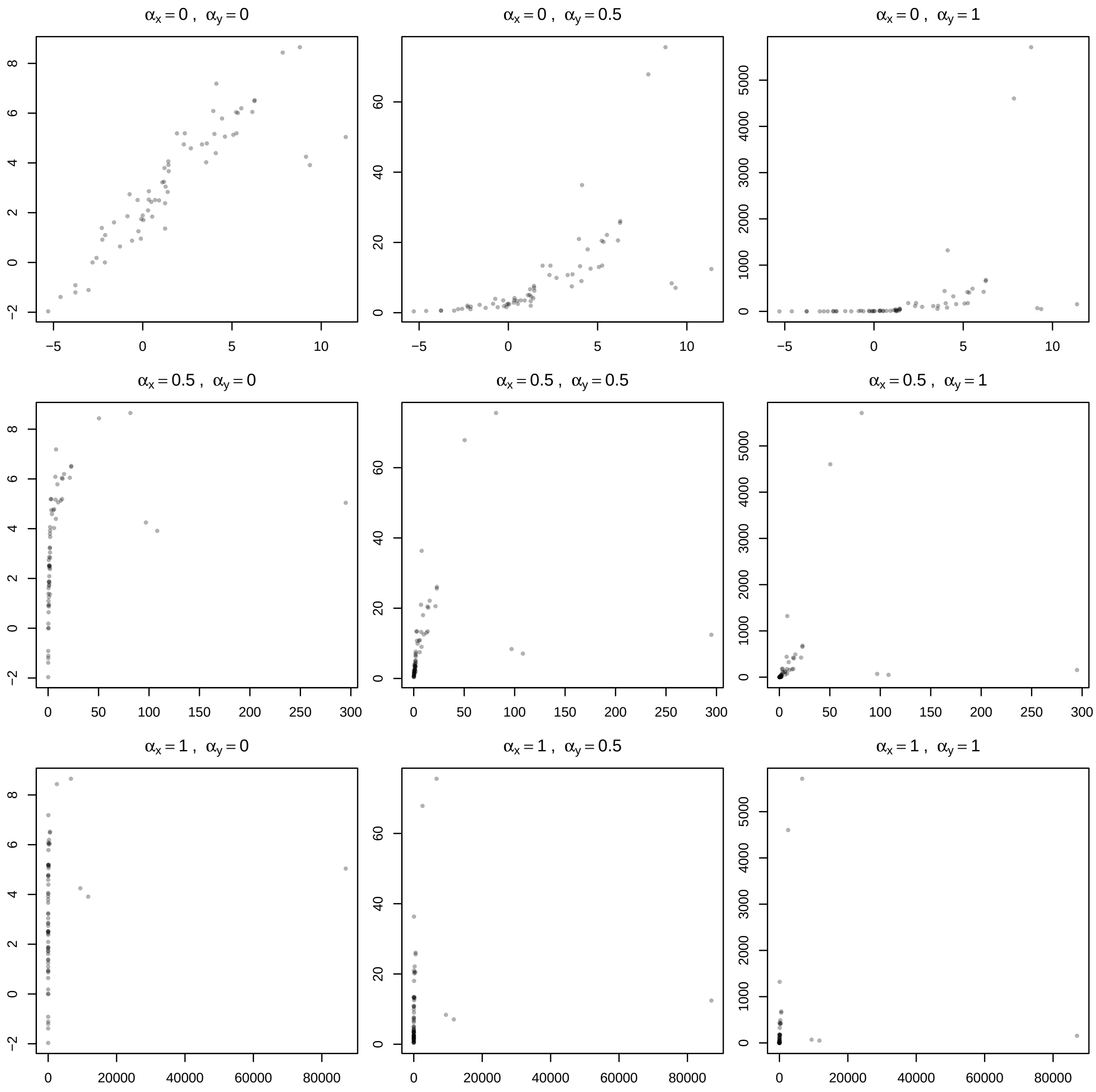
- Given the scatterplot above, can you see any relationship between body weight and brain weight?

- Both variates are very skewed, so a power transformation might help – which direction should we go with the powers?

- Let's look at the power-transformed data under several different transformations (choices of the α's):

```
powerfun <- function(y, alpha) {
    if (sum(y <= 0) > 0)
        stop("y must be positive")
    if (alpha == 0)
        log(y) else if (alpha > 0) {
        y^alpha
    } else -y^alpha
}

par(mfrow = c(3, 3), mar = 2.5 * c(1, 1, 1, 0.1))
a = rep(c(0, 1/2, 1), each = 3)
b = rep(c(0, 1/2, 1), times = 3)

for (i in 1:9) {
    plot(powerfun(Animals2$body, a[i]), powerfun(Animals2$brain, b[i]), pch = 19,
        cex = 0.5, col = adjustcolor("black", alpha = 0.3), xlab = "", ylab = "",
```

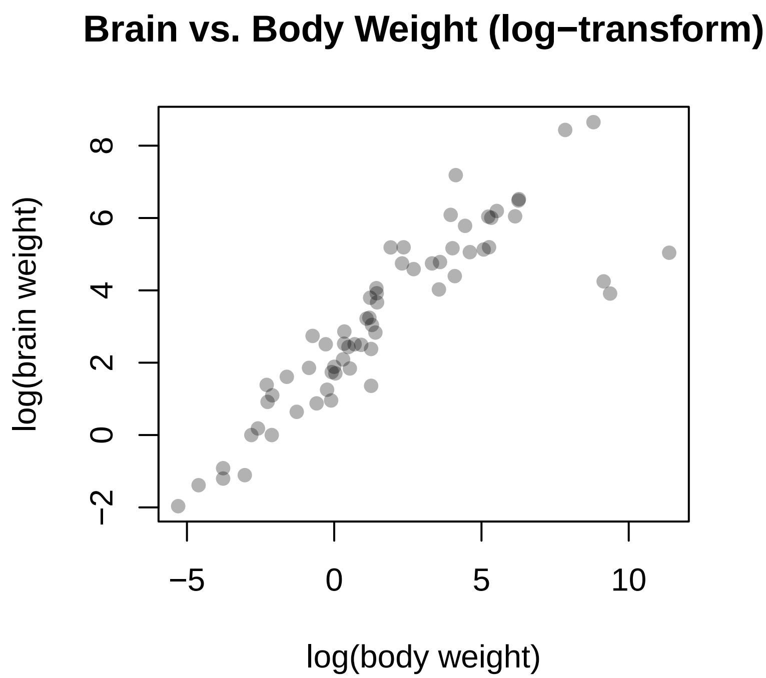
```
        main = bquote(alpha[x] == .(a[i]) ~ ", " ~ alpha[y] == .(b[i])))
    # main = paste('alpha_x = ', a[i] ,' & alpha_y=', b[i] ) )
}
```



- Do our "ladder rules" for power transformations seem applicable here?

- Which transformation do you prefer?

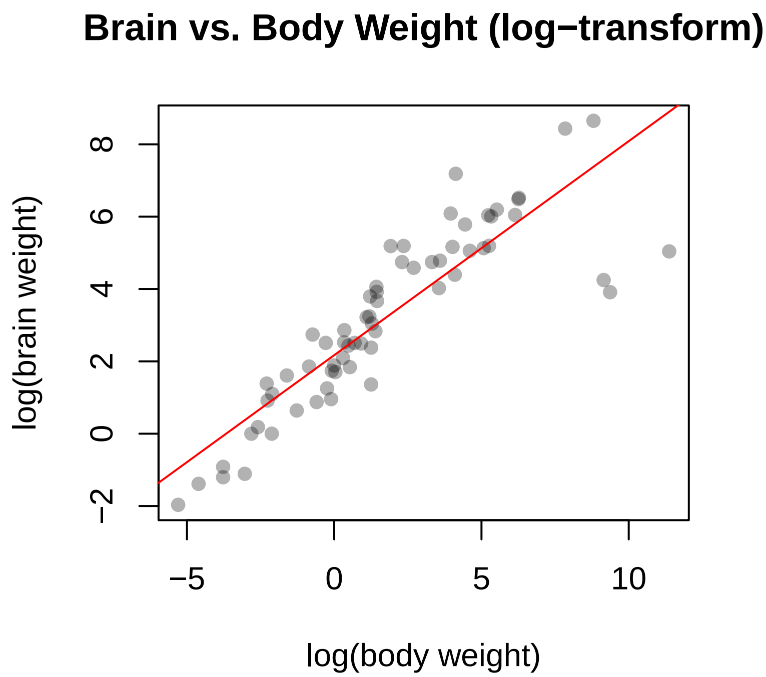
```r
plot(x = log(Animals2$body), y = log(Animals2$brain), main = "Brain vs. Body Weight (log-transform)",
    xlab = "log(body weight)", ylab = "log(brain weight)", pch = 16, col = adjustcolor("black",
        alpha.f = 0.3))
```



There appear to be 3 units that do not behave as the others do. It is likely that these will have a large **influence** on the fitted regression line. Let's fit the model and study the effect of these extreme units.

- Calculate and plot the least squares (LS) line of best fit for the transformed variates:

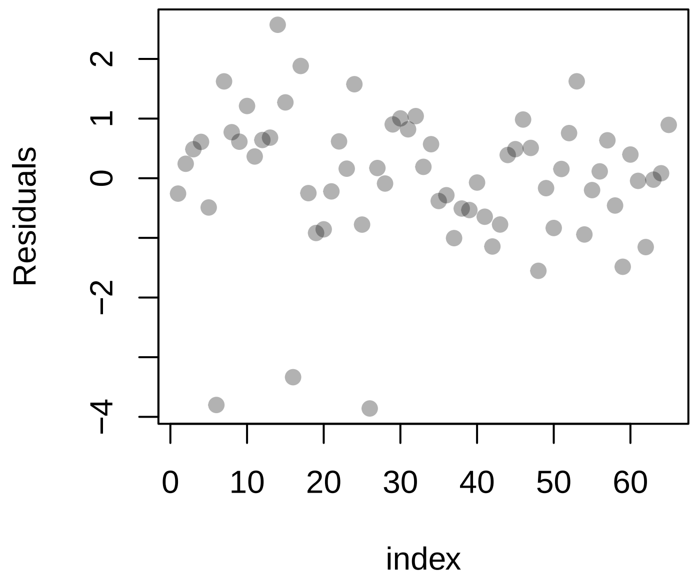
```r
plot(x = log(Animals2$body), y = log(Animals2$brain), main = "Brain vs. Body Weight (log-transform)",
    xlab = "log(body weight)", ylab = "log(brain weight)", pch = 16, col = adjustcolor("black",
        alpha.f = 0.3))
abline(lm(log(Animals2$brain) ~ log(Animals2$body)), col = "red")
```



- Does the LS regression line do a good job at representing this relationship?

- Let's analyze the residuals

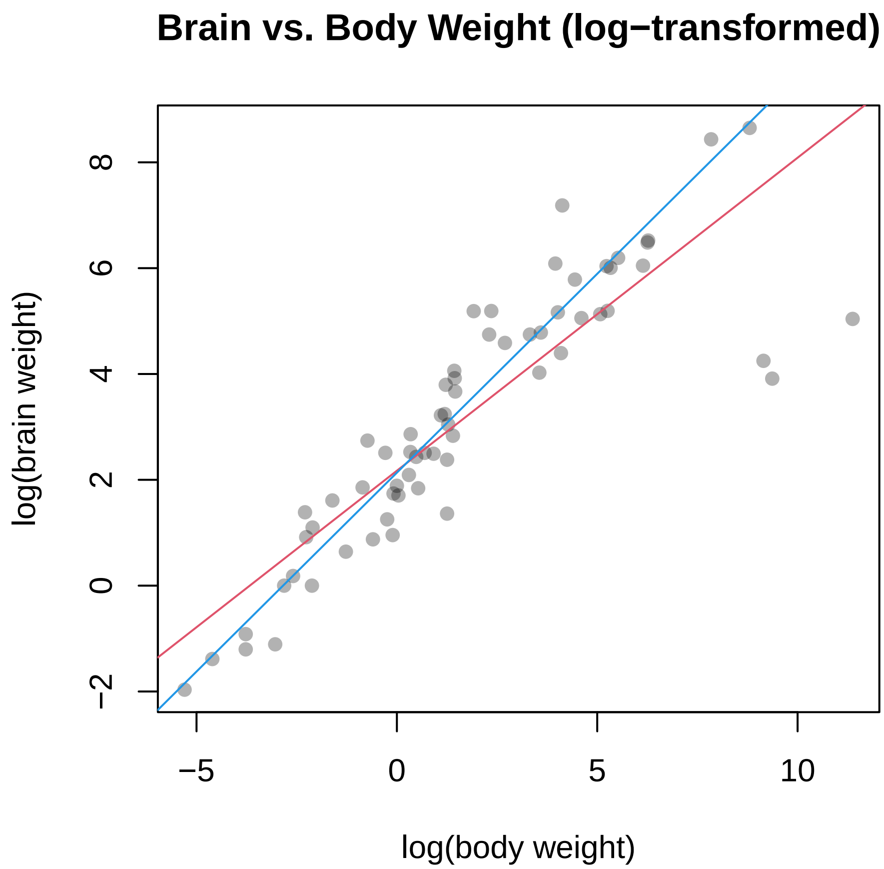
```
mod = lm(log(Animals2$brain) ~ log(Animals2$body))

plot(residuals(mod), col = adjustcolor("black", alpha = 0.3), ylab = "Residuals",
    xlab = "index", pch = 19)
```



- These confirm that there are three units that are somehow different from the others.
- We can either
  - remove the units, or
  - assign weights to the observations according to their variation, or
  - use a method to find the regression line which is *robust* to potential outliers.

- Let's begin by removing them:
  - `Red line`: LS regression using all the data
  - `Blue line`: LS regression while removing those three `outlier` points

```
mod = lm(log(Animals2$brain) ~ log(Animals2$body))

plot(x = log(Animals2$body), y = log(Animals2$brain), main = "Brain vs. Body Weight (log-transformed)",
    xlab = "log(body weight)", ylab = "log(brain weight)", pch = 16, col = adjustcolor("black",
        alpha.f = 0.3))

abline(mod, col = 2)

indx = which(log(Animals2$body) > 9)
W = rep(1, 65)
W[indx] = 0
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = W), col = 4)
```



- Although we have not formally calculated an influence statistic, the difference between the red and blue lines is a visual representation of the influence those three units have on the LS line.
- Which line seems like a better representation of the population?

So what are these units? Why are they so different from the others?

```
indx <- which(log(Animals2$body) > 9)
indx
```

```
## [1]  6 16 26
```

```
Animals2[indx, ]
```

```
##                 body brain
## Dipliodocus    11700  50.0
## Triceratops     9400  70.0
## Brachiosaurus 87000 154.5
```

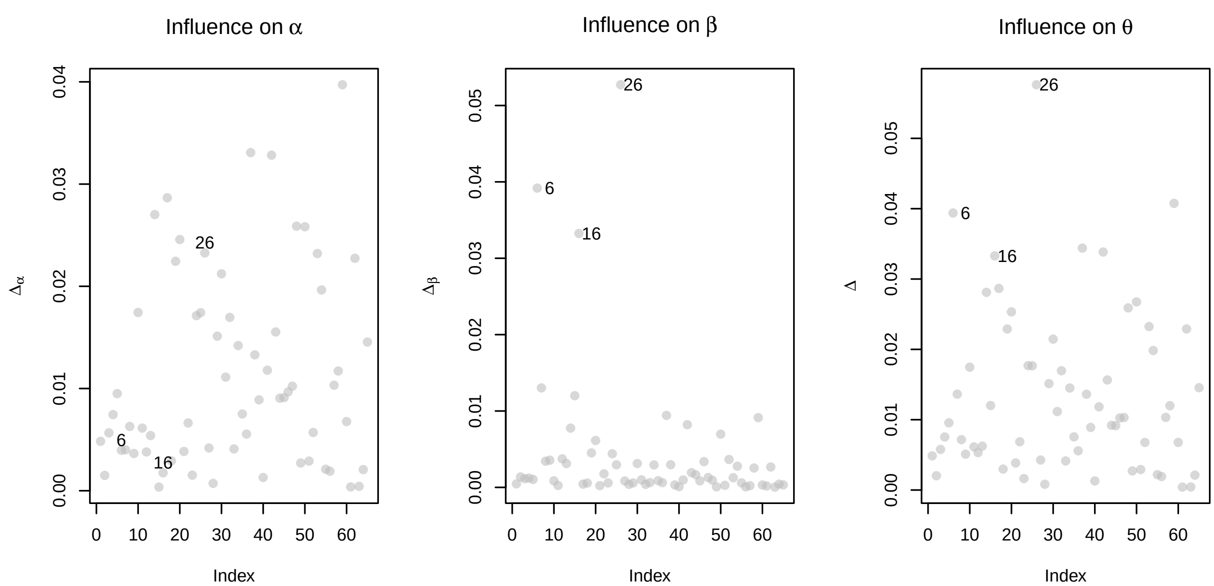
- These three data points are dinosaurs!

- The rest of the data-set are land mammals.
- Relative to other land mammals, dinosaurs have significantly larger bodies relative to their brain weights!

- Let us look at the influence of the points on the slope and the intercept of the LS regression line.

```
model.pop <- lm(log(Animals2$brain) ~ log(Animals2$body))
theta.hat <- model.pop$coef

N = nrow(Animals2)
delta = matrix(0, nrow = N, ncol = 2)

for (i in 1:N) {
    temp.model = lm(brain ~ body, data = log(Animals2[-i, ]))
    delta[i, ] = abs(theta.hat - temp.model$coef)
}

par(mfrow = c(1, 3))
plot(delta[, 1], ylab = bquote(Delta[alpha]), main = bquote("Influence on" ~
    alpha), pch = 19, col = adjustcolor("grey", 0.6))

obs = c(6, 16, 26)
```

```r
text(obs, delta[obs, 1] + 0.001, obs)

plot(delta[, 2], ylab = bquote(Delta[beta]), main = bquote("Influence on" ~
    beta), pch = 19, col = adjustcolor("grey", 0.6))

obs = c(6, 16, 26)
text(obs + 3, delta[obs, 2], obs)

delta2 = apply(X = delta, MARGIN = 1, FUN = function(z) {
    sqrt(sum(z^2))
})
plot(delta2, ylab = bquote(Delta), main = bquote("Influence on" ~ theta), pch = 19,
    col = adjustcolor("grey", 0.6))

text(obs + 3, delta2[obs], obs)
```



- Note these units (dinosaurs) have a large influential on the slope but not on the intercept.

Rather than deleting the problematic units, let's next consider giving these units less weight.

Weighted Linear Regression

In Weighted Least Squares (WLS), the fitted line minimizes the following objective function

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta)\,\in\,\mathbb{R}^2}{\operatorname{argmin}} \sum_{u\,\in\,\mathcal{P}} w_u \big[y_u - \alpha - \beta(x_u - c)\big]^2$$

- It is assumed the weights w_u are known but, as we will see, the residuals from an ordinary LS regression model can help us determine sensible values.

- As in ordinary LS regression a choice for c needs to be made. In this setting it is common to either set $c = 0$ or define c to be the weighted average of the x_u values:

$$c = \bar{x}_w = \frac{\sum_{u \in \mathcal{P}} w_u x_u}{\sum_{u \in \mathcal{P}} w_u}$$

- Given the values of the w_u's and c, we determine $\widehat{\boldsymbol{\theta}} = \left(\widehat{\alpha}, \widehat{\beta}\right)$ by taking derivatives of $\rho(\boldsymbol{\theta}; \mathcal{P})$ with respect to each parameter and then setting the resulting gradient equal to zero and solving the system of equations.

$$\sum_{u \in \mathcal{P}} w_u \left[y_u - \alpha - \beta(x_u - c)\right] \begin{bmatrix} 1 \\ \\ x_u - c \end{bmatrix} = \mathbf{0}$$

- Doing so yields the following estimates (**show this**):

$$\widehat{\alpha} = \bar{y}_w - \widehat{\beta}\,(\bar{x}_w - c) \qquad \text{and} \qquad \widehat{\beta} = \frac{\sum_{u \in \mathcal{P}} w_u (x_u - \bar{x}_w)(y_u - \bar{y}_w)}{\sum_{u \in \mathcal{P}} w_u (x_u - \bar{x}_w)^2}$$

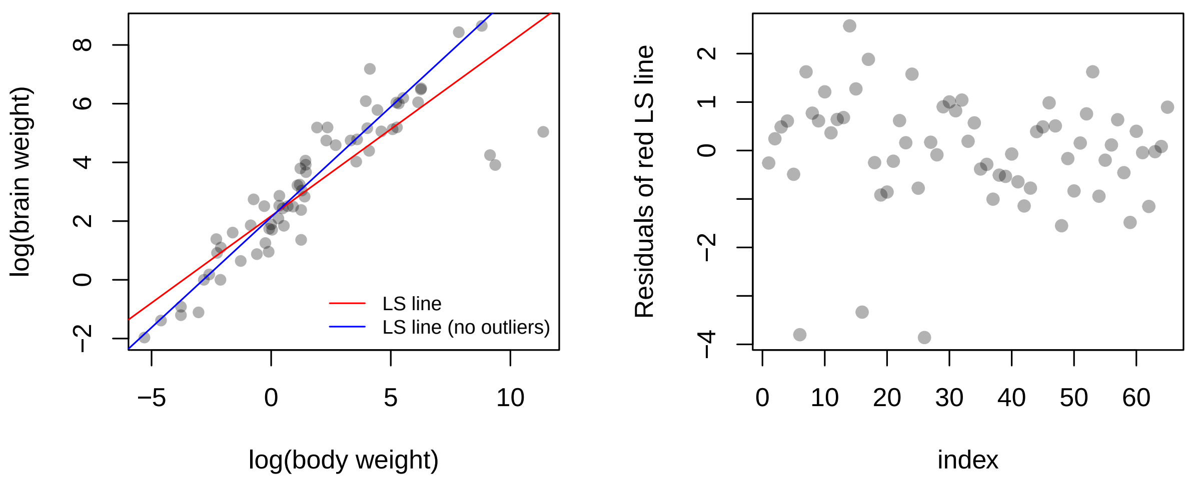
  where $\bar{y}_w$ and $\bar{x}_w$ are respectively the weighted averages of the y and x values.

Animals Example

We've seen previously that working with the log-transformed variates `log(brain weight)` and `log(body weight)` improved our ability to model the relationship between brain and body weight. We also saw that there were 3 unusual units (they were dinosaurs) that behaved differently than the others.

Recall:

```
par(mfrow = c(1, 2))
plot(x = log(Animals2$body), y = log(Animals2$brain), main = "", xlab = "log(body weight)",
    ylab = "log(brain weight)", pch = 16, col = adjustcolor("black", alpha.f = 0.3))

abline(lm(log(Animals2$brain) ~ log(Animals2$body)), col = "red")
W = rep(1, 65)
W[which(log(Animals2$body) > 9)] = 0
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = W), col = "blue")

legend("bottomright", legend = c("LS line", "LS line (no outliers)"), col = c("red",
    "blue"), cex = 0.75, bty = "n", lty = 1)

plot(residuals(lm(log(Animals2$brain) ~ log(Animals2$body))), col = adjustcolor("black",
    alpha = 0.3), ylab = "Residuals of red LS line", xlab = "index", pch = 19)
```



Rather than removing these units from the dataset and fitting an ordinary LS regression model, here we will fit a WLS regression model which includes all of the data, but that differentially weights the data points.

- Notice that some units have different residual variation than others.
- The residual standard deviations for both the outliers and the remaining points are:

```
res = residuals(lm(log(Animals2$brain) ~ log(Animals2$body)))

outlier.sd = sqrt(sum(res[indx]^2)/(length(indx)))
remaining.sd = sqrt(sum(res[-indx]^2)/(length(res[-indx])))

c(outlier.sd, remaining.sd)
```

```
## [1] 3.6723598 0.8626269
```

The relative variation (the ratio between these) is

```
remaining.sd/outlier.sd
```

```
## [1] 0.2348972
```

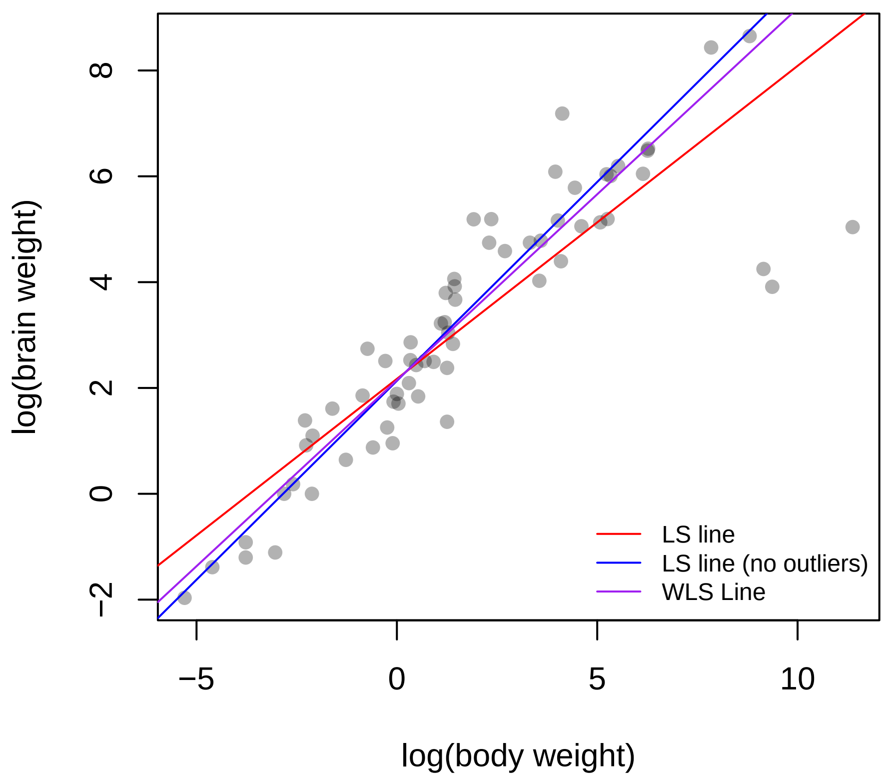
- Since the dinosaurs have larger residual variation we should give them less weight.
- Since the relative variation is roughly 0.2349 we could consider letting

$$w_u = \begin{cases} 1 & \text{for mammals} \\ 0.2349 & \text{for dinosaurs} \end{cases}$$

and then solve

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta)\,\in\,\mathbb{R}^2}{\text{argmin}} \sum_{u\,\in\,\mathcal{P}} w_u \left[y_u - \alpha - \beta(x_u - c)\right]^2$$

```r
plot(x = log(Animals2$body), y = log(Animals2$brain), main = "", xlab = "log(body weight)",
    ylab = "log(brain weight)", pch = 16, col = adjustcolor("black", alpha.f = 0.3))

abline(lm(log(Animals2$brain) ~ log(Animals2$body)), col = "red")
W = rep(1, 65)
W[which(log(Animals2$body) > 9)] = 0
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = W), col = "blue")
wt = rep(1, 65)
wt[obs] = remaining.sd/outlier.sd
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = wt), col = "purple")

legend("bottomright", legend = c("LS line", "LS line (no outliers)", "WLS Line"),
    col = c("red", "blue", "purple"), cex = 0.75, bty = "n", lty = 1)
```



- Which line best represents the relationship in the population?

Both of the procedures for dealing with outliers discussed so far (deletion and re-weighting) have been very manual. It would be nice to have a more automatic procedure to do this.

Robust Regression

- In weighted least squares the objective function is modified **manually** to give influential units less weight in the objective function. These influential points are far from the least squares line. Then we constructed

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta)\,\in\,\mathbb{R}^2}{\operatorname{argmin}} \sum_{u\,\in\,\mathcal{P}} w_u \left[y_u - \alpha - \beta(x_u - c)\right]^2 \qquad \text{where} \qquad w_u = \begin{cases} 1 & \text{for mammals} \\ 0.2349 & \text{for dinosaurs} \end{cases}$$

- Robust regression has the same goal. That is, points that are far from the linear line should have less weight in the objective function. We can modify the least square objective function to be

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta)\,\in\,\mathbb{R}^2}{\operatorname{argmin}} \sum_{u\,\in\,\mathcal{P}} \rho\left(y_u - \alpha - \beta(x_u - c)\right)$$

  where different forms of the function $\rho(\cdot)$ give rise to different fitted lines:

  - Least Squares Regression: equal weight on every single unit

  $$\rho\left(y_u - \alpha - \beta(x_u - c)\right) = \left[y_u - \alpha - \beta(x_u - c)\right]^2$$

  - Weighted Least Squares Line: give less weight to units with LS residuals that are large (in magnitude)

  $$\rho\left(y_u - \alpha - \beta(x_u - c)\right) = w_u\left[y_u - \alpha - \beta(x_u - c)\right]^2$$

- In robust regression we modify the loss function $\rho(y_u - \alpha - \beta\ (x_u - c))$ so that

  - it gives lower weight than least squares to units with large residuals (i.e., u such that $|r_u| >> 0$), and that
  - it is quadratic near 0 and hence behaves similarly to LS for units with small residuals (i.e., u such that $|r_u| \approx 0$)

- The **Huber Loss Function** achieves these goals by combining the the quadratic and absolute value functions:

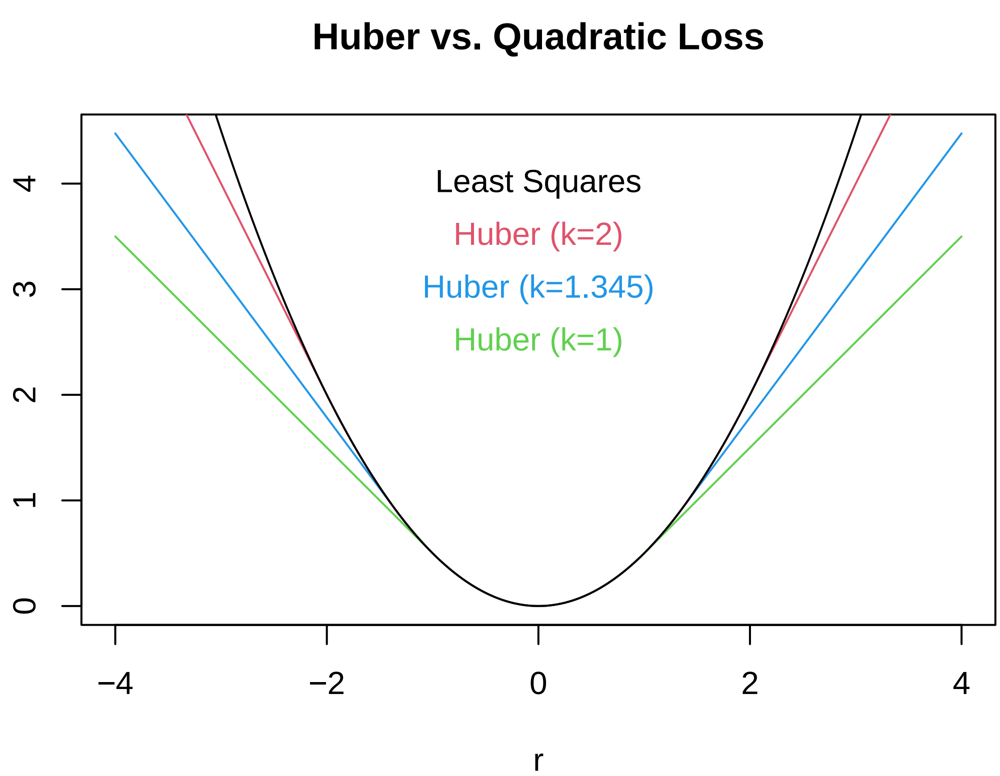
$$\rho_k(r) = \begin{cases} \frac{1}{2}r^2 & \text{for } |r| \leq k \\ k\ |r| - \frac{1}{2}k^2 & \text{for } |r| > k \end{cases}$$

Here is a general-purpose R implementation of the Huber Function:

```
huber.fn <- function(r, k) {
    val = r^2/2
    subr = abs(r) > k
    val[subr] = k * (abs(r[subr]) - k/2)
    return(val)
}
```

And here is a plot of the Huber Function for different choices of the threshold k:

```r
r = seq(-4, 4, 0.01)
plot(r, huber.fn(r, k = 1.345), type = "l", col = 4, main = "Huber vs. Quadratic Loss",
    ylab = "")
lines(r, huber.fn(r, k = 2), type = "l", col = 2)
lines(r, huber.fn(r, k = 1), type = "l", col = 3)
lines(r, r^2/2, type = "l")
text(0, 4, "Least Squares", bty = "n")
text(0, 3.5, "Huber (k=2)", bty = "n", col = 2)
text(0, 3, "Huber (k=1.345)", bty = "n", col = 4)
text(0, 2.5, "Huber (k=1)", bty = "n", col = 3)
```



- An attribute (e.g., regression coefficients) based on this function will be affected by the scale of r, and so…
  - we might let $k = cS$ where S is a (possibly robust) measure of scale.

- In practice it is common
  - to satisfy a theoretical balance between efficiency and resistance to outliers, we set $k \approx 1.345S$.
  - Other common choices include $k = 1.5$ or 2.

- Note: as k increases, the robust regression with Huber function imposes a larger penalty on larger residuals, hence approaches the least squares fit.

- Another form of **robust regression** involves defining the loss function in terms of **least absolute deviations (LAD)**:

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta) \, \in \, \mathbb{R}^2}{\operatorname{argmin}} \sum_{u \, \in \, \mathcal{P}} |y_u - \alpha - \beta(x_u - c)|$$

  - However, in both Huber and LAD-based regression the attribute $(\widehat{\alpha}, \widehat{\beta})$ cannot be solved for in closed form.

- When the attribute of interest is

$$(\widehat{\alpha}, \widehat{\beta}) = \underset{(\alpha,\beta) \, \in \, \mathbb{R}^2}{\operatorname{argmin}} \sum_{u \, \in \, \mathcal{P}} \rho\left(y_u - \alpha - \beta(x_u - c)\right)$$

  but the definition of $\rho(\cdot)$ precludes straightforward calculation, we consider the following optimization methods:

  - Gradient descent
  - Newton-Raphson
  - Iteratively reweighted least-squares

- The algorithms above are employed very generally to handle attributes defined implicitly as

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \, \in \, \boldsymbol{\Theta}}{\operatorname{argmin}} \, \rho(\boldsymbol{\theta}; \mathcal{P})$$

Animals Example

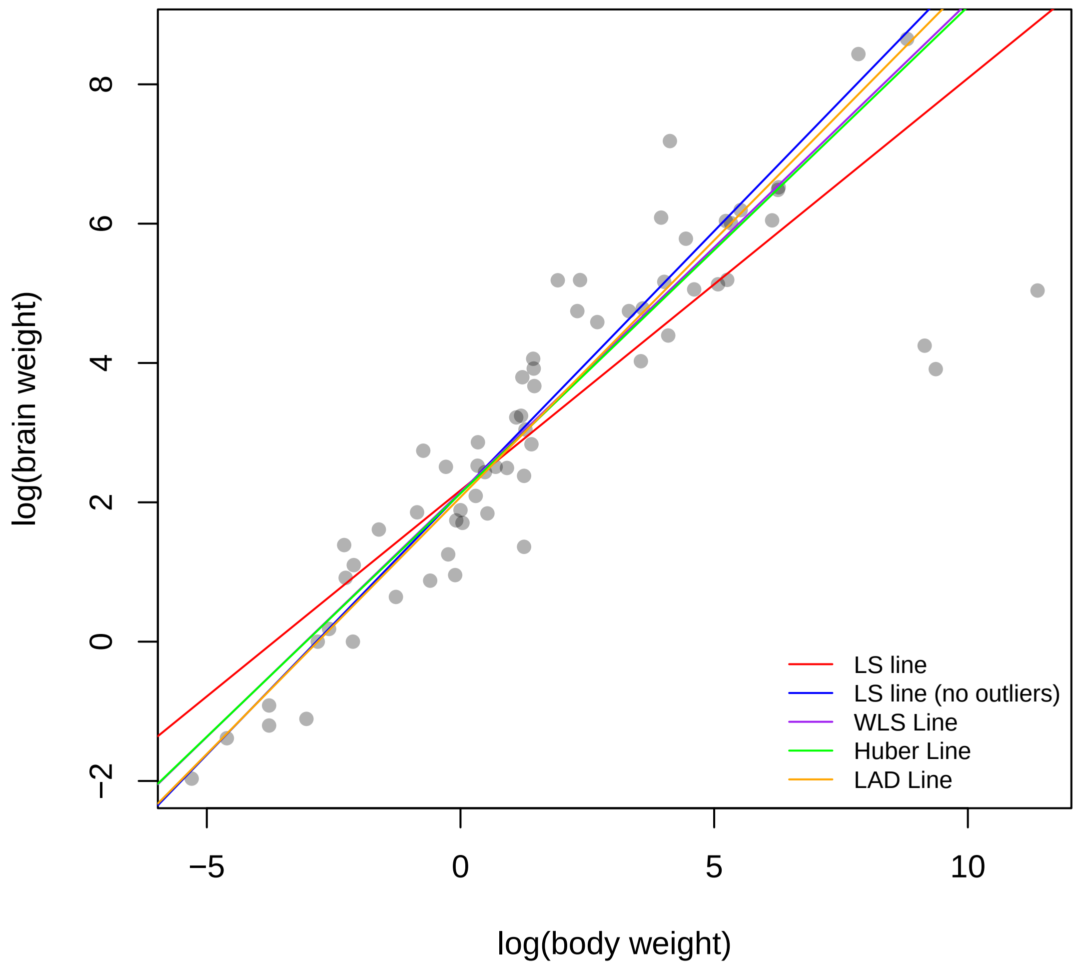
Without going through the details of the aforementioned optimization methods, here we simply implement Huber and LAD-based linear regression in the context of the `Animals` data to illustrate their robustness to the dinosaurs. We do so using existing `R` functions without explaining their implementation (yet).

```r
plot(x = log(Animals2$body), y = log(Animals2$brain), main = "", xlab = "log(body weight)",
    ylab = "log(brain weight)", pch = 16, col = adjustcolor("black", alpha.f = 0.3))

abline(lm(log(Animals2$brain) ~ log(Animals2$body)), col = "red")
W = rep(1, 65)
W[which(log(Animals2$body) > 9)] = 0
```

```r
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = W), col = "blue")
wt = rep(1, 65)
wt[obs] = remaining.sd/outlier.sd
abline(lm(log(Animals2$brain) ~ log(Animals2$body), weights = wt), col = "purple")

library(MASS)
abline(rlm(log(Animals2$brain) ~ log(Animals2$body), psi = "psi.huber"), col = "green")

library(L1pack)
```

```
## Loading required package: fastmatrix
```

```r
abline(lad(log(Animals2$brain) ~ log(Animals2$body)), col = "orange")

legend("bottomright", legend = c("LS line", "LS line (no outliers)", "WLS Line",
    "Huber Line", "LAD Line"), col = c("red", "blue", "purple", "green", "orange"),
    cex = 0.75, bty = "n", lty = 1)
```



- Which line best represents the relationship in the population?

Summary

- We reviewed power transformations and influence by examining the body and brain weight of a variety of animals.
- We used the `Animals` data to motivate the use of weighted and/or robust regression.
- In both cases we saw that changing the loss function can impact the influence of points that deviate from the linear line.

Exercises

From the Implicit Attribute exercises try problem 3 parts (a), (b), (c) and (d).

2.3.2a Gradient Descent

Overview

- Here we introduce gradient descent and show how it may be used to numerically determine implicitly defined attributes.
- We begin by considering the gradient descent algorithm in general. Then we examine how to implement the algorithm in R.
- We then consider two examples.

Introduction

Given an implicitly defined attribute of interest $\boldsymbol{\theta}$, the goal is to construct an **iterative procedure** which produces a sequence of **iterates**

$$\widehat{\boldsymbol{\theta}}_0,\ \widehat{\boldsymbol{\theta}}_1,\ \widehat{\boldsymbol{\theta}}_2,\ \ldots,\ \widehat{\boldsymbol{\theta}}_i,\ \widehat{\boldsymbol{\theta}}_{i+1},\ \ldots$$

such that this sequence converges to the solution

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}\,\in\,\boldsymbol{\Theta}}{\operatorname{argmin}}\, \rho(\boldsymbol{\theta};\mathcal{P}).$$

- Ideally, each iterate is closer to the solution $\widehat{\boldsymbol{\theta}}$ than the one before it.

- Assuming we are at some point $\widehat{\boldsymbol{\theta}}_i$ on the surface of $\rho(\boldsymbol{\theta};\mathcal{P})$, we achieve this by moving away from $\widehat{\boldsymbol{\theta}}_i$ in a direction which takes us to a point on the surface that is lower than at our current position, i.e., $\rho(\widehat{\boldsymbol{\theta}}_{i+1};\mathcal{P}) < \rho(\widehat{\boldsymbol{\theta}}_i;\mathcal{P})$.

- The direction of this movement is defined by the **gradient** of the surface.

Direction and Step Size

If $\rho(\boldsymbol{\theta};\mathcal{P})$ is a **differentiable** function of $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_k) \in \mathbb{R}^k$ then we can calculate the **gradient** for any value of $\boldsymbol{\theta}$:

$$\boldsymbol{g} = \boldsymbol{g}(\boldsymbol{\theta}) = \nabla\rho(\boldsymbol{\theta};\mathcal{P}) = \begin{bmatrix} \frac{\partial\rho(\boldsymbol{\theta};\mathcal{P})}{\partial\theta_1} \\ \frac{\partial\rho(\boldsymbol{\theta};\mathcal{P})}{\partial\theta_2} \\ \vdots \\ \frac{\partial\rho(\boldsymbol{\theta};\mathcal{P})}{\partial\theta_k} \end{bmatrix}$$

- Note that we will typically distinguish among the gradient calculations at each iteration
- At iteration i, when $\widehat{\boldsymbol{\theta}}_i$ is our best guess at the solution, we denote the gradient by $\boldsymbol{g}_i = \boldsymbol{g}(\widehat{\boldsymbol{\theta}}_i)$.

By definition, the normalized gradient

$$\boldsymbol{d}_i = \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$

provides the **direction** in which $\rho(\boldsymbol{\theta};\mathcal{P})$ increases or decreases fastest. (Recall that $\|\boldsymbol{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_k^2}$ for $\boldsymbol{x} \in \mathbb{R}^k$). In particular:

- $\boldsymbol{d}_i$ indicates the direction of **steepest ascent**, and
- $-\boldsymbol{d}_i$ indicates the direction of **steepest descent**

We iterate and obtain a new estimate of $\boldsymbol{\theta}$ by

- moving in the direction of $-\boldsymbol{d}_i$ and
- taking a step of size $\lambda_i > 0$

$$\widehat{\boldsymbol{\theta}}_{i+1} = \widehat{\boldsymbol{\theta}}_i - \lambda_i \ \boldsymbol{d}_i.$$

Note that the step size λ at each iteration can been chosen in a variety of ways:

1. We could choose a fixed value for all i such as $\lambda_i = 0.1$

2. We could define a fixed sequence such as $\lambda_i = 0.1 + 1/i$

3. We could perform a **line search** and algorithmically choose the value of λ_i that minimizes

$$\rho\left(\widehat{\boldsymbol{\theta}}_i - \lambda_i \ \boldsymbol{d}_i\right)$$

   In other words, which step size in the direction $-\boldsymbol{d}_i$ away from $\widehat{\boldsymbol{\theta}}_i$, minimizes $\rho\left(\widehat{\boldsymbol{\theta}}_{i+1}; \mathcal{P}\right)$.

The Gradient Descent Algorithm

Given some initial value $\widehat{\boldsymbol{\theta}}_0$

1. **Initialize** ; $i \leftarrow 0$;

2. LOOP:
   a. **Gradient**:
   $$\boldsymbol{g}_i = \nabla\rho(\boldsymbol{\theta}; \mathcal{P})\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$
   b. Gradient **direction**:
   $$\boldsymbol{d}_i \leftarrow \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$
   c. **Line search**: Find the step size $\widehat{\lambda}_i$
   $$\widehat{\lambda}_i = \arg\min_{\lambda>0} \rho\left(\widehat{\boldsymbol{\theta}}_i - \lambda \ \boldsymbol{d}_i\right)$$
   d. **Update** the iterate:
   $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \widehat{\lambda}_i \ \boldsymbol{d}_i$$
   e. **Converged?**
   if the iterates are not changing, then **Return**
   else $i \leftarrow i + 1$ and repeat LOOP.

3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

R code for Gradient Descent

```
gradientDescent <- function(theta = 0, rhoFn, gradientFn, lineSearchFn, testConvergenceFn,
    maxIterations = 100, tolerance = 1e-06, relative = FALSE, lambdaStepsize = 0.01,
    lambdaMax = 0.5) {

    converged <- FALSE
    i <- 0

    while (!converged & i <= maxIterations) {
        g <- gradientFn(theta)  ## gradient
        glength <- sqrt(sum(g^2))  ## gradient direction
        if (glength > 0)
            d <- g/glength

        lambda <- lineSearchFn(theta, rhoFn, d, lambdaStepsize = lambdaStepsize,
            lambdaMax = lambdaMax)

        thetaNew <- theta - lambda * d
        converged <- testConvergenceFn(thetaNew, theta, tolerance = tolerance,
            relative = relative)
        theta <- thetaNew
        i <- i + 1
    }

    ## Return last value and whether converged or not
    list(theta = theta, converged = converged, iteration = i, fnValue = rhoFn(theta))
}
```

Remarks:

- Notice the gradient descent algorithm was defined even though none of the functions it requires exist!
  - These will need to be implemented for each particular function $\rho(\boldsymbol{\theta}; \mathcal{P})$ we wish to minimize.

- For production code we would do more error checking
  - We are avoiding this here to better illustrate the algorithm.

R code for Line Search and Test of Convergence

- Here we consider a simple grid search to find the step size.

```
### line searching could be done as a simple grid search
gridLineSearch <- function(theta, rhoFn, d, lambdaStepsize = 0.01, lambdaMax = 1) {
    ## grid of lambda values to search
    lambdas <- seq(from = 0, by = lambdaStepsize, to = lambdaMax)
```

```r
    ## line search
    rhoVals <- sapply(lambdas, function(lambda) {
        rhoFn(theta - lambda * d)
    })
    ## Return the lambda that gave the minimum
    lambdas[which.min(rhoVals)]
}
```

- We stop when two iterates are sufficiently close to one another, where "sufficiently" depends on a tolerance ϵ. That is,

$$\left\|\widehat{\boldsymbol{\theta}}_{i+1}-\widehat{\boldsymbol{\theta}}_{i}\right\|_{1}<\epsilon$$

  where $\|\cdot\|_1$ is the L_1 norm defined by $\|\mathbf{z}\|_1 = \sum_{j=1}^{k}|z_j|$ where $\mathbf{z}$ is a vector with dimension k.

- We could also measure this on a relative scale:

$$\frac{\left\|\widehat{\boldsymbol{\theta}}_{i+1}-\widehat{\boldsymbol{\theta}}_{i}\right\|_{1}}{\left\|\widehat{\boldsymbol{\theta}}_{i}\right\|_{1}}<\epsilon$$

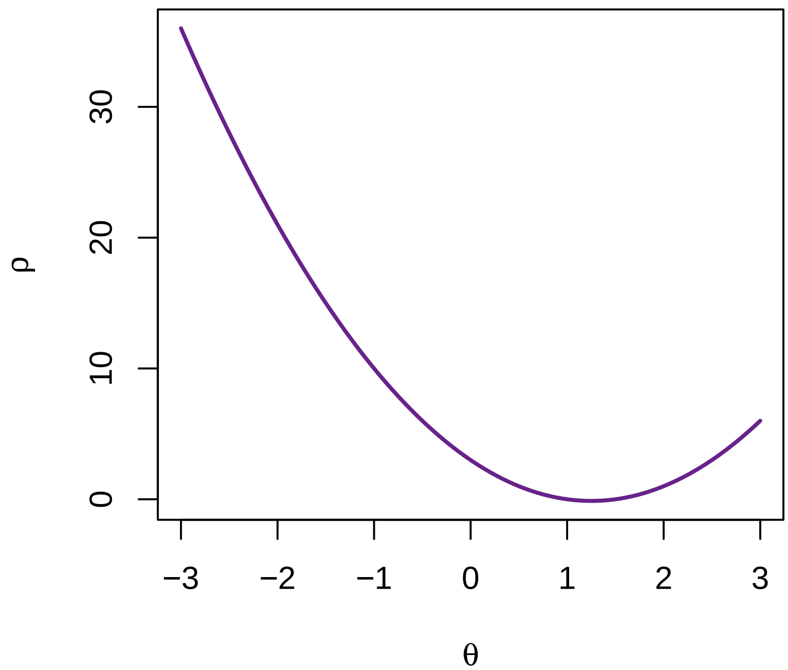
- **Remark:** We could change the L_1 norm to any other distance metric.

```r
### Where testCovergence might be (relative or absolute)
testConvergence <- function(thetaNew, thetaOld, tolerance = 1e-10, relative = FALSE) {
    sum(abs(thetaNew - thetaOld)) < if (relative)
        tolerance * sum(abs(thetaOld)) else tolerance
}
```

Example 1 (one-dimensional quadratic function)

Suppose for example we are interested in finding the value of θ which minimizes

$$\rho(\theta) = 2\theta^2 - 5\theta + 3$$



The gradient is

$$g = \rho'(\theta) = 4\theta - 5$$

and we only need to write the corresponding `R` functions

```
rho <- function(theta) {
    2 * theta^2 - 5 * theta + 3
}
g <- function(theta) {
    4 * theta - 5
}
```

Then perform Gradient Descent using our `gradientDescent` function to find the value $\widehat{\theta}$ which minimizes $\rho(\theta)$.

```
gradientDescent(rhoFn = rho, gradientFn = g, lineSearchFn = gridLineSearch,
    testConvergenceFn = testConvergence)
```

```
## $theta
## [1] 1.25
##
## $converged
## [1] TRUE
##
## $iteration
```

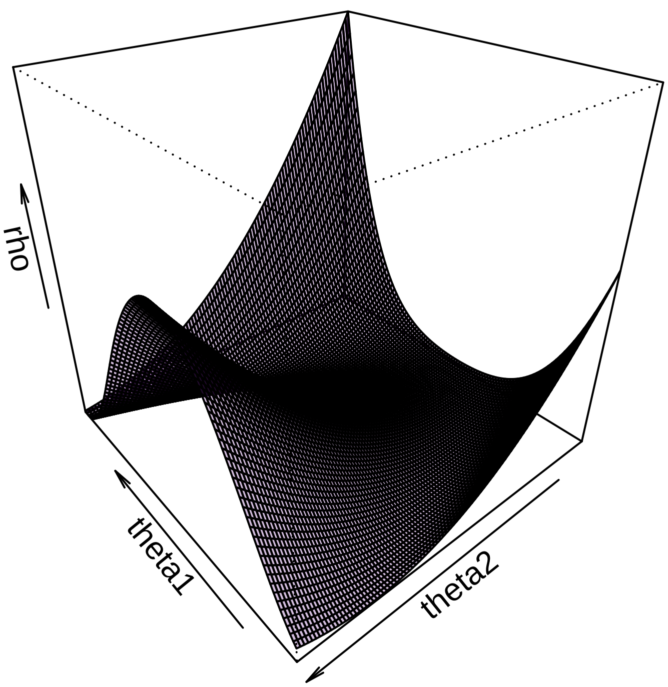
```
## [1] 4
##
## $fnValue
## [1] -0.125
```

Example 2 (two-dimensional Rosenbrock function)

We want to find $\boldsymbol{\theta} = (\theta_1, \theta_2)'$ which minimizes the following Rosenbrock function

$$\rho(\boldsymbol{\theta}) = (1 - \theta_1)^2 + 100\left(\theta_2 - \theta_1^2\right)^2$$

which is a **non-convex** function, usually used in performance test problems for optimization algorithms. A plot of this function is shown below for $\theta_1 \in [-1.5, 1.8]$ and $\theta_2 \in [-0.5, 3.0]$.



The gradient for this function is

$$\boldsymbol{g} = \nabla\rho(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial\rho}{\partial\theta_1} \\ \frac{\partial\rho}{\partial\theta_2} \end{bmatrix} = \begin{bmatrix} 400\theta_1^3 - 400\theta_1\theta_2 + 2\theta_1 - 2 \\ -200\theta_1^2 + 200\theta_2 \end{bmatrix}$$

To use our `gradientDescent` function let's define $\rho(\boldsymbol{\theta})$ and $\nabla\rho(\boldsymbol{\theta})$ in `R`:

```
rho <- function(theta) {
    (1 - theta[1])^2 + 100 * (theta[2] - theta[1]^2)^2
}
g <- function(theta) {
    c(400 * theta[1]^3 - 400 * theta[1] * theta[2] + 2 * theta[1] - 2, -200 *
        theta[1]^2 + 200 * theta[2])
}
```

Use the code below to find $\widehat{\theta}$ which minimizes $\rho(\boldsymbol{\theta})$.

```
gradientDescent(theta = c(0, 0), rhoFn = rho, gradientFn = g, lineSearchFn = gridLineSearch,
    lambdaStepsize = 1e-04, testConvergenceFn = testConvergence, maxIterations = 1000)
```

```
## $theta
## [1] 0.9940877 0.9882748
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 254
##
## $fnValue
## [1] 3.537011e-05
```

Summary

- We covered a general gradient descent algorithm and how to implement it in R.
  - This included the line search and the test of convergence.
- We looked at two examples.

Exercises

From the Implicit Attribute exercises try problem 4. For continued learning try problems 5 and 11 (but not part f).

2.3.2b Gradient Descent Regression Examples

Overview

- Here we consider two more examples that illustrate the utility of gradient descent.
  - We consider least squares regression and robust regression.
- We also discuss an important issue in R: we avoid reliance on the global environment by using functions that make functions.

The Gradient Descent Algorithm (Review)

Given some initial value $\widehat{\boldsymbol{\theta}}_0$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:

   a. **Gradient**:
   $$\boldsymbol{g}_i = \nabla\rho(\boldsymbol{\theta};\mathcal{P})\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$

   b. Gradient **direction**:
   $$\boldsymbol{d}_i \leftarrow \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$

   c. **Line search**: Find the step size $\widehat{\lambda}_i$
   $$\widehat{\lambda}_i = \arg\min_{\lambda>0} \rho\left(\widehat{\boldsymbol{\theta}}_i - \lambda\ \boldsymbol{d}_i\right)$$

   d. **Update** the iterate:
   $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \widehat{\lambda}_i\ \boldsymbol{d}_i$$

   e. **Converged?**
   if the iterates are not changing, then **Return**
   else $i \leftarrow i+1$ and repeat LOOP.

3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

Example 3 (Least Squares Regression)

Suppose we have a population $\mathcal{P}$ which consists of pairs (x_u, y_u) and we wish to fit the following linear regression model to these data

$$y_u = \alpha + \beta(x_u - \bar{x}) + r_u$$

The attribute of interest is $\boldsymbol{\theta} = (\alpha, \beta)^T$ and the objective function we need to minimize is

$$\rho(\boldsymbol{\theta}) = \rho(\alpha, \beta) = \sum_{u \in \mathcal{P}} (y_u - \alpha - \beta(x_u - \bar{x}))^2$$

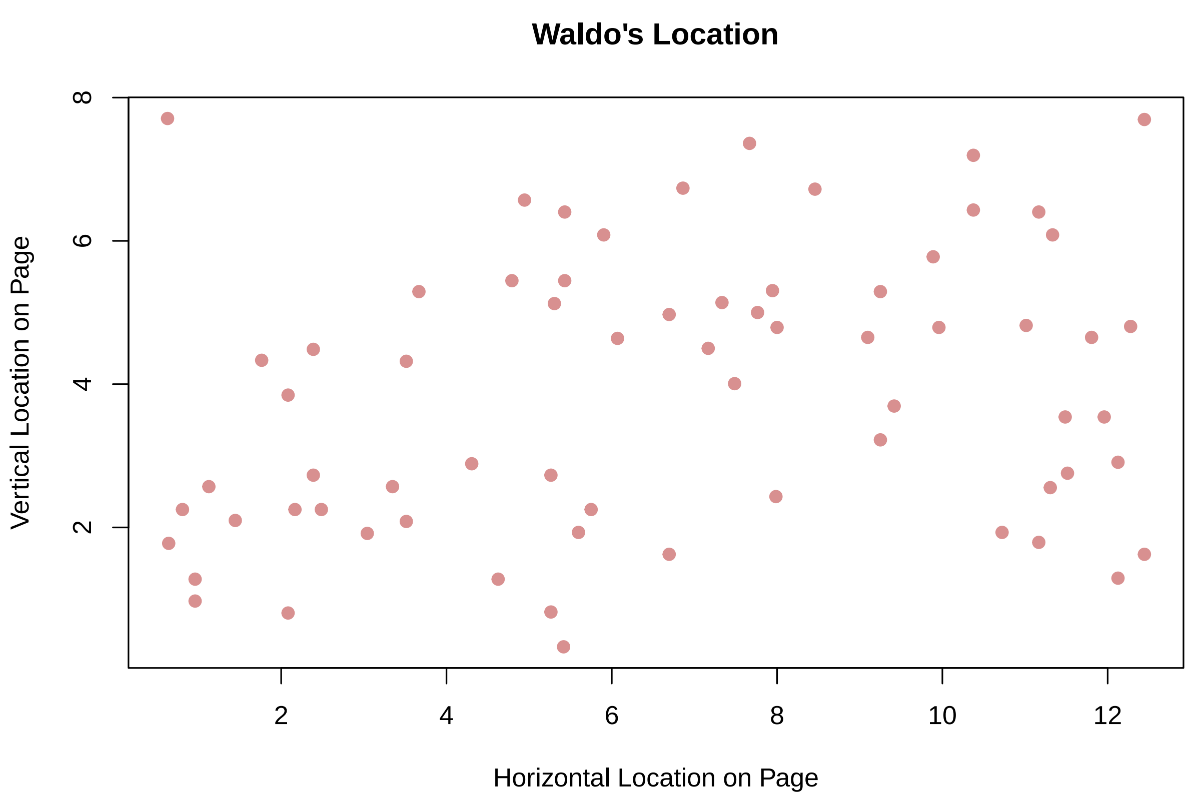
The corresponding gradient is

$$\boldsymbol{g} = \nabla \rho(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial \rho}{\partial \alpha} \\ \frac{\partial \rho}{\partial \beta} \end{bmatrix} = \begin{bmatrix} \sum_{u \in \mathcal{P}} -2\left(y_u - \alpha - \beta(x_u - \bar{x})\right) \\ \sum_{u \in \mathcal{P}} -2\left(y_u - \alpha - \beta(x_u - \bar{x})\right)(x_u - \bar{x}) \end{bmatrix}$$

Where's Waldo?

For this example we will take $\mathcal{P}$ to be the `Where's Waldo?` population for which x and y respectively represent the horizontal and vertical locations of `Waldo` in a given page in a given book.

```r
waldo <- read.csv("../../Data/wheres-waldo-locations.csv", header = TRUE)
plot(waldo$X, waldo$Y, pch = 19, col = adjustcolor("firebrick", 0.5), main = "Waldo's Location",
    xlab = "Horizontal Location on Page", ylab = "Vertical Location on Page")
```



As with the previous two examples we need to write the corresponding `R` functions `rho` and `gradient`. However, unlike the previous two examples, these functions will now rely on data in addition to $\boldsymbol{\theta}$.

```
rho <- function(theta) {
    alpha <- theta[1]
    beta <- theta[2]
    ## Note that we are accessing waldo from the globalEnv
    y <- waldo$Y
    x <- waldo$X
    xbar <- mean(x)
    sum((y - alpha - beta * (x - xbar))^2)
}

gradient <- function(theta) {
    alpha <- theta[1]
    beta <- theta[2]
    ## Note that we are accessing waldo from the globalEnv
    y <- waldo$Y
    x <- waldo$X
    xbar <- mean(x)
    g <- -2 * c(sum(y - alpha - beta * (x - xbar)), sum((y - alpha - beta *
        (x - xbar)) * (x - xbar)))
    # Return g
    g
}
```

Remarks:

- In the R code above *global* variables like `waldo$X` and `waldo$Y` appear inside the functions `rho(...)` and `gradient(...)`.
- This will work provided `waldo` is available in the *global environment* when `rho(...)` and `gradient(...)` are called. This may not be reliable and so should be avoided in general.

INTERLUDE: Avoiding Reliance on the Global Environment

- An obvious solution, and one which generally works, is to pass the needed variables (here `x` and `y`) to the relevant functions as arguments.
  - Unfortunately, for our purposes, this would clutter up the code somewhat and necessitate rewriting the nice and *very general* `gradientDescent` function.
  - A better way to proceed, which still keeps clutter to a minimum, is to write functions which will return the appropriate functions.

- Functions containing their own data environment are called **closures**.
  - Every function has a local environment where variables may be defined; this is the closure of the function.
  - Functions also have access to the environment in which they were created (that's why functions can access values in the global environment).

- We exploit this property by creating functions that themselves define and return a function.
  - The interior function is enclosed within the function that created it, hence the word closure.
  - The returned function (or closure) has access to any variables defined within the enclosing function that defined it.

- Encapsulation of data within a function is an important and powerful construct.

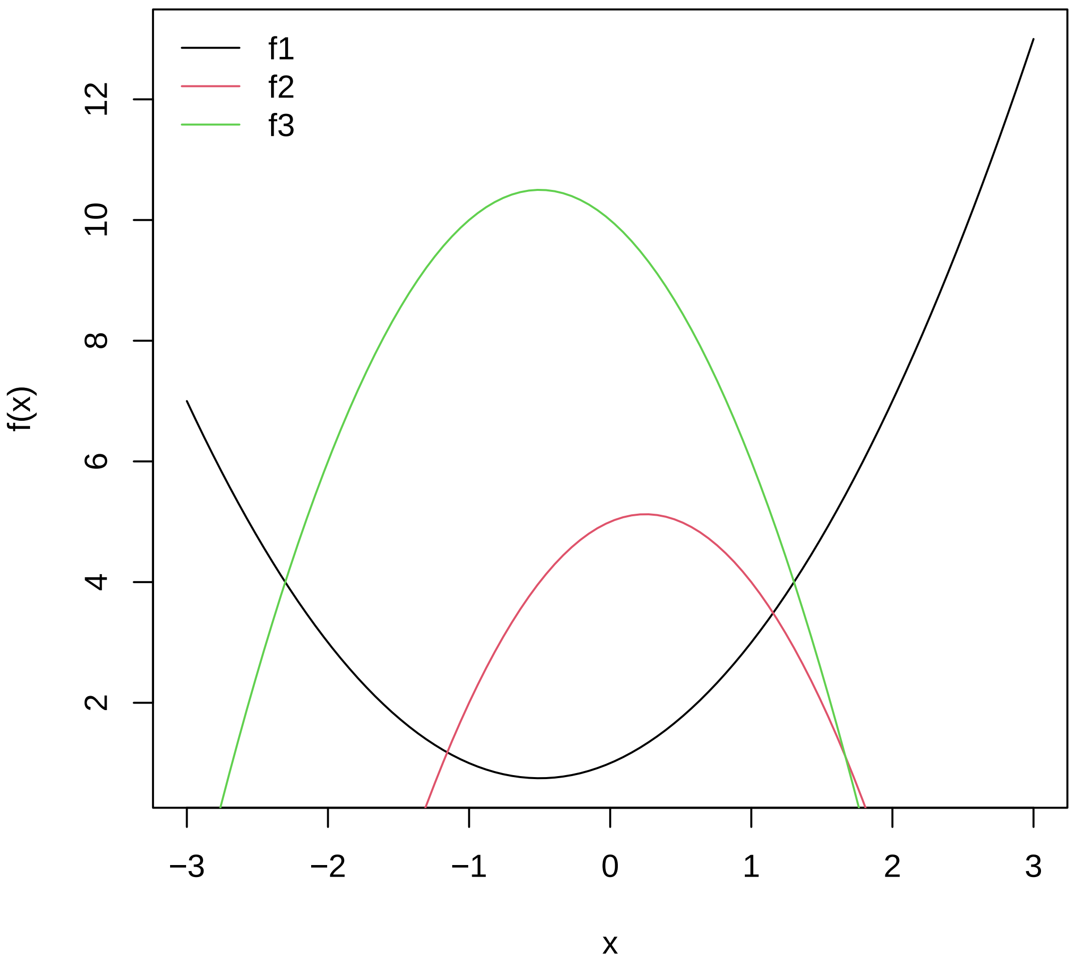
Factory Functions: Functions that make Functions

Here is a simple example of a function that defines and returns a quadratic function.

```
createQuadratic <- function(a, b, c) {
    ## Return this function
    function(x) {
        fx = a * x^2 + b * x + c
        return(fx)
    }
}
```

Here is our function-creating-function in action:

```
x = seq(-3, 3, length.out = 100)

f1 = createQuadratic(a = 1, b = 1, c = 1)
f2 = createQuadratic(a = -2, b = 1, c = 5)
f3 = createQuadratic(a = -2, b = -2, c = 10)

plot(x, f1(x), type = "l", ylab = "f(x)")
lines(x, f2(x), col = 2)
lines(x, f3(x), col = 3)

legend("topleft", lty = 1, col = 1:3, legend = c("f1", "f2", "f3"), bty = "n")
```



In a similar fashion we can define functions that will take in the data (`x` and `y`) and return appropriate `rho` and `gradient` functions for least squares regression with any data:

```
createLeastSquaresRho <- function(x, y) {
    ## local variable
    xbar <- mean(x)
    ## Return this function
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        sum((y - alpha - beta * (x - xbar))^2)
    }
}
```

```r
### We now get the rho function for the waldo data
rho <- createLeastSquaresRho(waldo$X, waldo$Y)

### Similarly for the gradient function
createLeastSquaresGradient <- function(x, y) {
    ## local variables
    xbar <- mean(x)
    ybar <- mean(y)
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        -2 * c(sum(y - alpha - beta * (x - xbar)), sum((y - alpha - beta * (x -
            xbar)) * (x - xbar)))
    }
}
gradient <- createLeastSquaresGradient(waldo$X, waldo$Y)
```

Back to Waldo

The functions `rho` and `gradient` can now be passed as arguments to `gradientDescent(...)` to find the value $\widehat{\boldsymbol{\theta}} = \left(\widehat{\alpha}, \widehat{\beta}\right)^T$ that minimizes

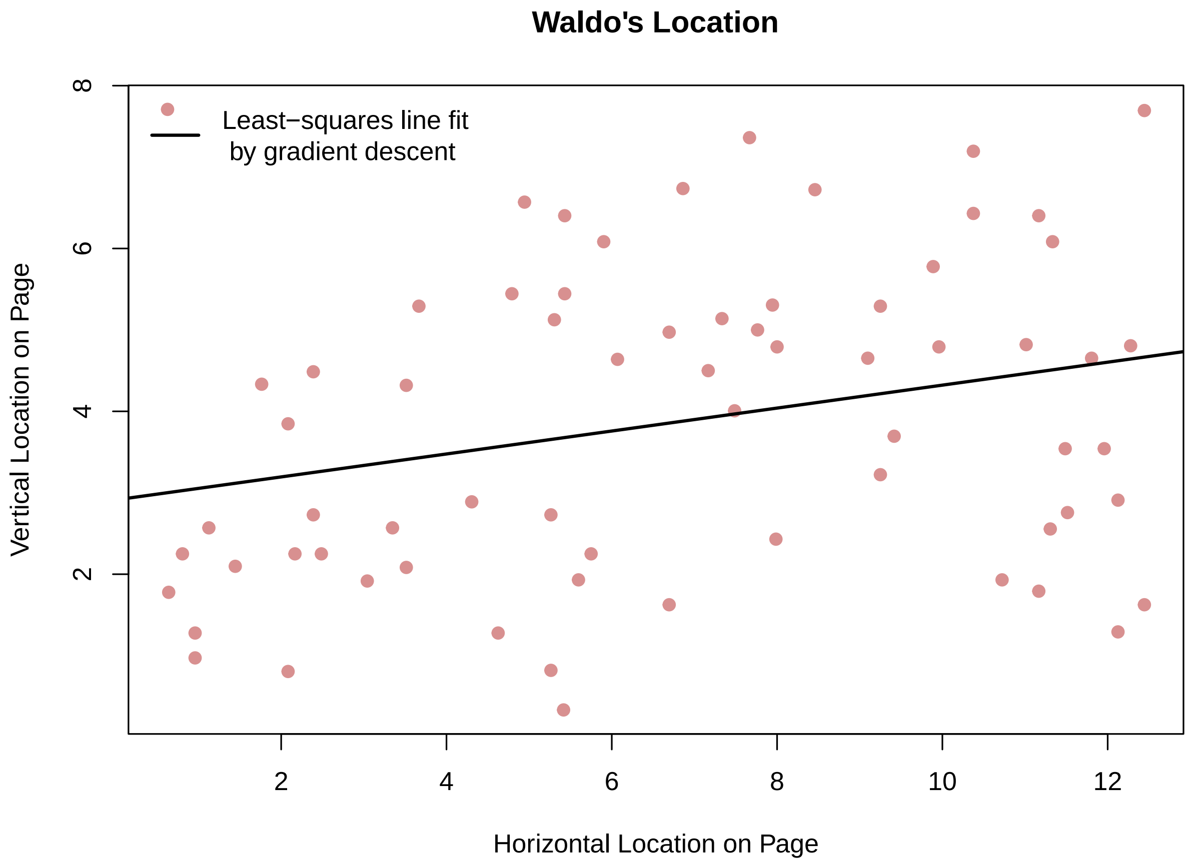
$$\rho(\boldsymbol{\theta}) = \rho(\alpha, \beta) = \sum_{u \in \mathcal{P}} \left(y_u - \alpha - \beta(x_u - \bar{x})\right)^2$$

```r
result <- gradientDescent(theta = c(0, 0), rhoFn = rho, gradientFn = gradient,
    lineSearchFn = gridLineSearch, testConvergenceFn = testConvergence)
print(result)
```

```
## $theta
## [1] 3.857138 0.140886
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 29
##
## $fnValue
## [1] 233.7737
```

And we can plot the resulting line for the `Where's Waldo?` population:

```r
plot(waldo$X, waldo$Y, pch = 19, col = adjustcolor("firebrick", 0.5), main = "Waldo's Location",
    xlab = "Horizontal Location on Page", ylab = "Vertical Location on Page")
abline(result$theta[1] - result$theta[2] * mean(waldo$X), result$theta[2], lwd = 2)
legend("topleft", lty = 1, col = 1, lwd = 2, legend = "Least-squares line fit \n by gradient descent",
    bty = "n")
```



Why do we care about this line?

Example 4 (Robust Regression)

As in the previous example, let us suppose we have a population $\mathcal{P}$ which consists of pairs (x_u, y_u) and we wish to fit the following linear regression model to these data

$$y_u = \alpha + \beta(x_u - \bar{x}) + r_u$$

The attribute of interest is again $\boldsymbol{\theta} = (\alpha, \beta)^T$ but now let's assume we wish to perform **robust regression** in which case the objective function to be minimized is

$$\rho(\boldsymbol{\theta}) = \rho(\alpha, \beta) = \sum_{u \in \mathcal{P}} \rho_k \left(y_u - \alpha - \beta(x_u - \bar{x}) \right) = \sum_{u \in \mathcal{P}} \rho_k(r_u)$$

where $r_u = y_u - \alpha - \beta(x_u - \bar{x})$ and

$$\rho_k(r) = \begin{cases} \frac{1}{2} r^2 & \text{for } |r| \leq k \\ k\,|r| - \frac{1}{2}k^2 & \text{for } |r| > k \end{cases}$$

is the Huber function.

The corresponding gradient is

$$\boldsymbol{g} = \nabla\rho(\boldsymbol{\theta})$$

which can be decomposed using the Chain Rule as follows:

$$\nabla\rho(\boldsymbol{\theta}) = \frac{\partial}{\partial\boldsymbol{\theta}}\sum_{u\in\mathcal{P}}\rho_k(r_u) = \sum_{u\in\mathcal{P}}\frac{\partial\rho_k(r_u)}{\partial\boldsymbol{\theta}} = \sum_{u\in\mathcal{P}}\frac{\partial\rho_k(r_u)}{\partial r_u}\times\frac{\partial r_u}{\partial\boldsymbol{\theta}}$$

where

$$\frac{\partial\rho_k(r)}{\partial r} = \begin{cases} r & \text{for } |r| \le k \\ \text{sign}(r)\times k & \text{for } |r| > k \end{cases}$$

and

$$\frac{\partial r_u}{\partial\boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial r_u}{\partial\alpha} \\ \frac{\partial r_u}{\partial\beta} \end{bmatrix} = -\begin{bmatrix} 1 \\ x_u - \bar{x} \end{bmatrix}$$

Thus the gradient can be written as

$$\boldsymbol{g} = -\begin{bmatrix} \sum_{u\in\mathcal{P}}\frac{\partial\rho_k(r_u)}{\partial r_u} \\ \sum_{u\in\mathcal{P}}\frac{\partial\rho_k(r_u)}{\partial r_u}(x_u - \bar{x}) \end{bmatrix}$$

The following are `R` functions for the Huber function and its derivative as well as *factory* functions which, given data, create `rho(...)` and `gradient(...)` functions for simple linear robust regression.

The Huber function

```
huber.fn <- function(r, k) {
    val = r^2/2
    subr = abs(r) > k
    val[subr] = k * (abs(r[subr]) - k/2)
    return(val)
}
```

A function to create the objective function

```
createRobustHuberRho <- function(x, y, kval) {
    ## local variable
    xbar <- mean(x)
    ## Return this function
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        sum(huber.fn(y - alpha - beta * (x - xbar), k = kval))
    }
}
```

The derivative of the Huber function

```
huber.fn.prime <- function(r, k) {
    val = r
    subr = abs(r) > k
    val[subr] = k * sign(r[subr])
    return(val)
}
```

A function to create the gradient function

```
createRobustHuberGradient <- function(x, y, kval) {
    ## local variables
    xbar <- mean(x)
    ybar <- mean(y)
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        ru = y - alpha - beta * (x - xbar)
        rhok = huber.fn.prime(ru, k = kval)
        -1 * c(sum(rhok * 1), sum(rhok * (x - xbar)))
    }
}
```

Animals

Here we will use gradient descent with the functions defined above to perform robust regression on the `Animals` data that we've worked with previously. The necessary R code to do so is provided below:

```
data(Animals2, package = "robustbase")
rho <- createRobustHuberRho(x = log(Animals2$body), y = log(Animals2$brain),
    kval = 1.345)

gradient <- createRobustHuberGradient(x = log(Animals2$body), y = log(Animals2$brain),
    kval = 1.345)

result <- gradientDescent(theta = c(0, 0), rhoFn = rho, gradientFn = gradient,
    lineSearchFn = gridLineSearch, testConvergenceFn = testConvergence)

print(result)
```

```
## $theta
## [1] 3.3235437 0.6834746
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 20
##
## $fnValue
## [1] 31.20838
```

Let's compare these results to what we found previously (which relied on the `rlm` function in the `MASS` package):

```
library(MASS)
temp = rlm(log(Animals2$brain) ~ I(log(Animals2$body) - mean(log(Animals2$body))),
    psi = "psi.huber")
temp$coef
```

```
##                                        (Intercept)
##                                          3.3405534
## I(log(Animals2$body) - mean(log(Animals2$body)))
##                                          0.6985483
```

Summary

- We considered two more examples for gradient descent. In the robust regression example we were able to reproduce results from other R packages. As an exercise, compare our least squares linear regression results for the Waldo data with what is obtained when using the `lm` function.
- We learned how to make functions that make functions to avoid reliance on the global environment in R.

Exercises

From the Implicit Attribute exercises try problems 1, 9 and 10 (except part j). For continued learning try problem 3 (except part g), and 14.

2.3.2c Gradient Descent in Batches

Overview

- We explore variations of the gradient descent algorithm which can be applied when the objective function to be minimized has the following form:

$$\rho(\boldsymbol{\theta}, \mathcal{P}) = \sum_{u \in \mathcal{P}} \rho(\boldsymbol{\theta}; u)$$

- We see how to implement these algorithms in R.

The Gradient Descent Algorithm (Review)

Given some initial value $\widehat{\boldsymbol{\theta}}_0$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:

    a. **Gradient**:

    $$\boldsymbol{g}_i = \nabla \rho(\boldsymbol{\theta}; \mathcal{P})\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$

    b. Gradient **direction**:

    $$\boldsymbol{d}_i \leftarrow \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$

    c. **Line search**: Find the step size $\widehat{\lambda}_i$

    $$\widehat{\lambda}_i = \arg\min_{\lambda > 0} \quad \rho\left(\widehat{\boldsymbol{\theta}}_i - \lambda \; \boldsymbol{d}_i\right)$$

    d. **Update** the iterate:

    $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \widehat{\lambda}_i \; \boldsymbol{d}_i$$

    e. **Converged?**
    if the iterates are not changing, then **Return**
    else $i \leftarrow i + 1$ and repeat LOOP.

3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

Batch Gradient Descent

In practice, many of the objective functions minimized during statistical analyses have the following form:

$$\rho(\boldsymbol{\theta}, \mathcal{P}) = \sum_{u \in \mathcal{P}} \rho(\boldsymbol{\theta}; u)$$

in which case the gradient $\boldsymbol{g}$ can simply be written as the sum of the unit-specific contributions to the objective function:

$$\boldsymbol{g} = \boldsymbol{g}(\boldsymbol{\theta}) = \nabla \rho(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \nabla \rho(\boldsymbol{\theta}; u) \equiv \sum_{u \in \mathcal{P}} \boldsymbol{g}(\boldsymbol{\theta}; u)$$

Thus when $\rho(\cdot)$ is a sum over $u \in \mathcal{P}$:

- the gradient $\boldsymbol{g}$ is composed of N 'smaller' independent gradient calculations
- these individual gradient calculations $\boldsymbol{g}(\boldsymbol{\theta}; u)$ can be done in any order
  - this can be very handy when N is large and the individual gradients are expensive to calculate
  - we may wish to perform the gradient computations in several batches which are distributed across different machines
  - this is important in many "Big Data" applications

- In this situation the terms **batch gradient descent** and **gradient descent** are used interchangeably.
  - The appropriateness of the term **batch** becomes clear when we explicitly partition the population $\mathcal{P}$ into H non-overlapping groups (batches)

    $$\mathcal{P} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \cdots \cup \mathcal{B}_H$$

    each containing M_k units ($k = 1, 2, \ldots, H$):

    $$\boldsymbol{g} = \sum_{u \in \mathcal{P}} \boldsymbol{g}(\boldsymbol{\theta}; u) = \sum_{k=1}^{H} \sum_{u \in \mathcal{B}_k} \boldsymbol{g}(\boldsymbol{\theta}; u)$$

Batch gradient descent lends itself well to parallel computation. But what if the gradient calculations are sufficiently complex and even parallelization isn't fast enough?

Gradient Descent Using Subsets of the Population

When computing the gradient $\boldsymbol{g}$ is computationally expensive we could consider using only a subset of the available data – as opposed to using all of it.

- If the run time for batch gradient descent based on all N units is too long, consider *estimating* the gradient using just $M < N$ units

- In such situations we typically do not optimize for the step size λ and instead use a fixed step size $\lambda^\star$ (Why?)
  - $\lambda^\star$ is often referred to as the **learning rate**

- Two common approaches to do this are **batch-sequential** and **batch-stochastic** gradient descent
  - these approaches differ only in the manner in which the the subsets of size M are chosen.

Batch-Sequential Gradient Descent

- Suppose we can divide the population of size N to H batches of size M, i.e., $N = H \times M$ and $\mathcal{P} = \{\mathcal{B}_1, \ldots, \mathcal{B}_H\}$
- In this approach we sequentially move through the H batches and update our estimate $\widehat{\boldsymbol{\theta}}$ after *each* batch.
  - Note that this is different from ordinary batch gradient descent; in that case the gradients are still calculated in batches, but the $\widehat{\boldsymbol{\theta}}$ is only updated *after observing all batches.*

- If convergence takes more than H iterations then the batches are iteratively sequenced through until convergence.

- The **batch-sequential gradient descent algorithm** is the following:

Given some initial value $\widehat{\boldsymbol{\theta}}_0$ and a fixed step size $\lambda^\star$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:
   a. **Gradient**:
   $$\boldsymbol{g}_i = \nabla\rho(\boldsymbol{\theta}; \mathcal{B}_{i \bmod H})\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$
   b. Gradient **direction**:
   $$\boldsymbol{d}_i \leftarrow \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$
   c. **Update** the iterate:
   $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \lambda^\star\ \boldsymbol{d}_i$$
   d. **Converged?**
   if the iterates are not changing, then **Return**
   else $i \leftarrow i+1$ and repeat LOOP.
3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

Batch-Stochastic Gradient Descent

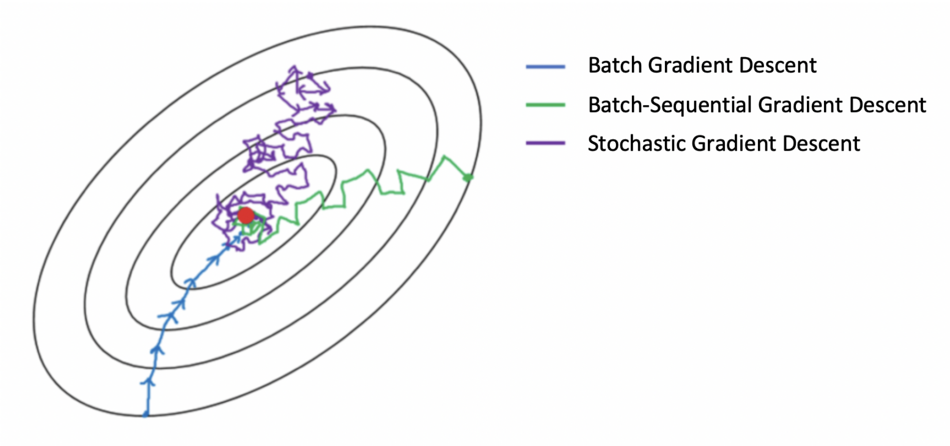
- In this approach, each iteration of the gradient is calculated from a sample (batch) $\mathcal{S}$ selected randomly from the population $\mathcal{P}$.
  - Like batch-sequential gradient descent, the estimate $\widehat{\boldsymbol{\theta}}$ is updated after *each* batch (sample). Denoting the sample size by M, note that setting $M = 1$ gives rise to what is often simply referred to as **stochastic gradient descent**.

- The **batch-stochastic gradient descent** algorithm is the following:

  Given some initial value $\widehat{\boldsymbol{\theta}}_0$ and a fixed step size $\lambda^\star$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:
   a. **Gradient**: Given a **new** random sample $\mathcal{S} \in \mathcal{P}$
   $$\boldsymbol{g}_i = \nabla\rho(\boldsymbol{\theta}; \mathcal{S})\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$
   b. Gradient **direction**:
   $$\boldsymbol{d}_i \leftarrow \frac{\boldsymbol{g}_i}{\|\boldsymbol{g}_i\|}$$
   c. **Update** the iterate:
   $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \lambda^\star\ \boldsymbol{d}_i$$
   d. **Converged?**
   if the iterates are not changing, then **Return**
   else $i \leftarrow i+1$ and repeat LOOP.
3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

Comparing the Algorithms



R code for Batch-Stochastic Gradient Descent

- Do we need to change any code in the `gradientDescent` function?

- We have to edit the `gradient` function. For example, here is a factory function that will create the batch-stochastic gradient function for robust regression with the Huber function.

```
createRobustHuberGradient.stochastic <- function(x, y, kval, M) {
    subset <- sample(x = length(x), size = M)
    x <- x[subset]
    y <- y[subset]
    xbar <- mean(x)
    ybar <- mean(y)
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        ru <- y - alpha - beta * (x - xbar)
        rhok <- huber.fn.prime(ru, k = kval)
        -1 * c(sum(rhok), sum((rhok * (x - xbar))))
    }
}
```

- And if we want to use fixed step sizes we need to write a function that does this

```
fixedLineSearch <- function(theta, rhoFn, g, lambdaStepsize = 0.01, lambdaMax = 1) {
    lambdaStepsize
}
```

- Let's put this together and perform batch-stochastic gradient descent when performing robust regression on the `Animals` data:

The following are `R` functions for the Huber function and its derivative as well as a *factory* function which, given data, creates `rho(...)`. Note that we have already defined the *factory* function that creates the `gradient(...)` function.

The Huber function

```
huber.fn <- function(r, k) {
    val = r^2/2
    subr = abs(r) > k
    val[subr] = k * (abs(r[subr]) - k/2)
    return(val)
}
```

The derivative of the Huber function

```
huber.fn.prime <- function(r, k) {
    val = r
    subr = abs(r) > k
    val[subr] = k * sign(r[subr])
    return(val)
}
```

A function to create the objective function

```
createRobustHuberRho <- function(x, y, kval) {
    ## local variable
    xbar <- mean(x)
    ## Return this function
    function(theta) {
        alpha <- theta[1]
        beta <- theta[2]
        sum(huber.fn(y - alpha - beta * (x - xbar), k = kval))
    }
}
```

Animals

Here we will use batch-stochastic gradient descent with the functions defined above to perform robust regression on the `Animals` data that we've worked with previously. The necessary R code to do so is provided below. Note that `relative=TRUE` is *required* in the test of convergence here (why?).

```
data(Animals2, package = "robustbase")

rho <- createRobustHuberRho(x = log(Animals2$body), y = log(Animals2$brain),
    kval = 1.345)

gradient.stochastic <- createRobustHuberGradient.stochastic(x = log(Animals2$body),
    y = log(Animals2$brain), kval = 1.345, M = 10)
set.seed(341)
```

```r
result2 <- gradientDescent(theta = c(1, 0), rhoFn = rho, gradientFn = gradient.stochastic,
    lineSearchFn = fixedLineSearch, testConvergenceFn = testConvergence, maxIterations = 10^5,
    tolerance = 0.001, lambdaStepsize = 0.01, relative = TRUE)

print(result2)
```

```
## $theta
## [1] 3.203477 0.406294
##
## $converged
## [1] FALSE
##
## $iteration
## [1] 100001
##
## $fnValue
## [1] 50.79372
```

Let's compare these results to what we found previously (which relied on the `rlm` function in the `MASS` package):

```r
library(MASS)
temp = rlm(log(Animals2$brain) ~ I(log(Animals2$body) - mean(log(Animals2$body))),
    psi = "psi.huber")
temp$coef
```

```
##                                      (Intercept)
##                                        3.3405534
## I(log(Animals2$body) - mean(log(Animals2$body)))
##                                        0.6985483
```

Exercise: examine the effect of changing the value of `M`. Any value `M>1` results in batch-stochastic gradient descent.

Summary

- We explored two variations of the gradient descent algorithm which can be applied when the objective function to be minimized has the following form:

$$\rho(\boldsymbol{\theta}, \mathcal{P}) = \sum_{u \in \mathcal{P}} \rho(\boldsymbol{\theta}; u)$$

- In **batch-sequential** gradient descent, we divide in the population into H batches. Then in each iteration perform one step of gradient descent using one of the H batches. Each iteration uses a different batch and we move through the population, batch-by-batch sequentially.
- In **batch-stochastic** gradient descent, at each iteration we randomly sample a batch and then perform one step of gradient descent using that batch (sample).

Exercises

From the Implicit Attribute exercises try problem 13.

2.3.3a Systems of Equations

Overview

- We have defined an attribute $a(\mathcal{P})$ to be a summary of the population $\mathcal{P}$.
- We discussed numeric and graphical attributes that we called **explicit** attributes because they are defined *explicitly* by the population variates.
- More recently we have been discussing **implicitly** defined attributes, which also summarize the population $\mathcal{P}$, but which are defined only *implicitly* by the population variates.
  - Such attributes are solutions to an optimization problem, which so far we have cast as a minimization problem.
- Here, we provide an alternative definition of implicitly defined attributes: **the solution to an equation or a set of equations**
  - We consider how to use Newton's Method to numerically solve such systems of equations.

We have defined an **implicit attribute $\boldsymbol{\theta}$** as the solution to an optimization problem. Until now we have cast such an optimization problem as the minimization of some appropriately defined objective function $\rho(\boldsymbol{\theta}; \mathcal{P})$:

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \boldsymbol{\Theta}}{\operatorname{argmin}} \, \rho(\boldsymbol{\theta}; \mathcal{P})$$

In all situations we saw that such a solution was obtained by solving (either algorithmically or in closed form) the equation

$$\nabla \rho(\boldsymbol{\theta}; \mathcal{P}) = \mathbf{0}$$

When the dimension of the vector valued attribute $\boldsymbol{\theta}$ is k, such an equation is actually a system of k independent equations with k unknowns. Thus an implicit attribute $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ can be defined slightly more generally as the solution to a system of equations

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \mathbf{0}$$

Thus we work directly with $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})$ which, in most cases, equals $\nabla \rho(\boldsymbol{\theta}; \mathcal{P})$.

Unsurpsingly, $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})$ is often defined as a sum over $u \in \mathcal{P}$:

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \boldsymbol{\psi}(\boldsymbol{\theta}; u)$$

in which case the attribute of interest $\widehat{\boldsymbol{\theta}}$ is the value $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ that solves

$$\sum_{u \in \mathcal{P}} \boldsymbol{\psi}(\boldsymbol{\theta}; u) = \mathbf{0}$$

Examples (scalar valued attributes)

Recall our familiar examples for implicitly defined scalar attributes θ

- Previously these were defined as solutions to minimization problems
- Now we define them as solutions to systems of equations

The average is the value of θ which solves

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} y_u - n\theta = 0$$

The weighted average is the value of θ which solves

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} w_u(y_u - \theta) = 0$$

The q^{th} quantile: is the smallest value of θ which solves

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \psi(\theta; u) = \sum_{u \in \mathcal{P}} \frac{1}{N} I\left(y_u \leq \theta\right) - q = 0$$

Examples (vector valued attributes)

Recall our interest in the vector valued attribute $\boldsymbol{\theta} = (\alpha, \beta)$ in the context of a simple linear regression. Different forms of linear regression arose by determining the vector $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})$ which was the solution to a variety of different systems of equations.

Least squares

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \sum_{u \in \mathcal{P}} \boldsymbol{\psi}(\boldsymbol{\theta}; u) = \sum_{u \in \mathcal{P}} (y_u - \alpha - \beta(x_u - c)) \begin{pmatrix} 1 \\ \\ x_u - c \end{pmatrix} = \mathbf{0}$$

Weighted least squares

$$\boldsymbol{\psi}(\boldsymbol{\theta};\mathcal{P}) = \sum_{u\in\mathcal{P}} \boldsymbol{\psi}(\boldsymbol{\theta};u) = \sum_{u\in\mathcal{P}} w_u(y_u - \alpha - \beta(x_u - c)) \begin{pmatrix} 1 \\ \\ x_u - c \end{pmatrix} = \mathbf{0}$$

Robust regression

$$\boldsymbol{\psi}(\boldsymbol{\theta};\mathcal{P}) = \sum_{u\in\mathcal{P}} \boldsymbol{\psi}(\boldsymbol{\theta};u) = \sum_{u\in\mathcal{P}} \rho_k'(y_u - \alpha - \beta(x_u - c)) \begin{pmatrix} 1 \\ \\ x_u - c \end{pmatrix} = \mathbf{0}$$

The class of methods used to find such solutions to systems of equations are generally referred to as **root finding** methods.

Newton's Method

Suppose we have a differentiable function $f(x)$ and we wish to find $x = x^\star$ which solves

$$f(x) = 0$$

Given an initial value x_0

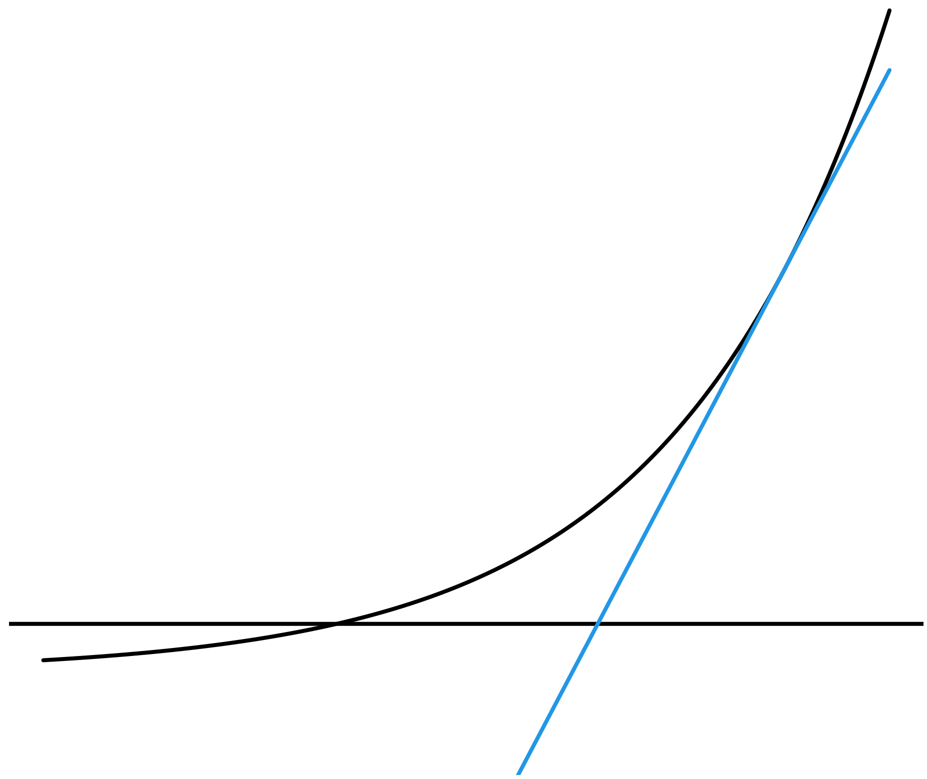
- we can use a linear function to approximate $f(x)$ in the vicinity of x_0

$$f(x) \approx f(x_0) + f'(x_0)\,(x - x_0)$$

- then we find the root of the linear approximation to iterate to the next value of x:

$$0 = f(x_0) + f'(x_0)\,(x - x_0) \qquad \Rightarrow \qquad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

- We continue in this way until x_{i+1} does not differ much from x_i



For our implicitly defined scalar attribute θ, we want to find $\theta \in \Theta$ such that

$$\psi\left(\theta; \mathcal{P}\right) = 0$$

- Given the current guess $\widehat{\theta} = \widehat{\theta}_i$ a first order approximation is

$$\psi(\theta; \mathcal{P}) \approx \psi(\widehat{\theta}_i; \mathcal{P}) + \psi'(\widehat{\theta}_i; \mathcal{P}) \times \left(\theta - \widehat{\theta}_i\right).$$

- Then the update is the root of the linear approximation and is given by

$$\widehat{\theta}_{i+1} = \widehat{\theta}_i - \frac{\psi(\widehat{\theta}_i; \mathcal{P})}{\psi'(\widehat{\theta}_i; \mathcal{P})}$$

The Newton Algorithm

Given an initial value $\widehat{\theta}_0$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:

   a. **Update** the iterate:
   $$\widehat{\theta}_{i+1} \leftarrow \widehat{\theta}_i - \frac{\psi(\widehat{\theta}_i; \mathcal{P})}{\psi'(\widehat{\theta}_i; \mathcal{P})}$$

   b. **Converged?**
   if the iterates are not changing, then **return**
   else $i \leftarrow i + 1$ and repeat LOOP.
3. **Return**: $\widehat{\theta} = \widehat{\theta}_i$

R Code for Newton's Method

Here is a basic `R` implementation of Newton's method:

```
Newton <- function(theta = 0,
                   psiFn, psiPrimeFn,
                   testConvergenceFn = testConvergence,
                   maxIterations = 100,   # maximum number of iterations
                   tolerance = 1E-6,      # parameters for the test
                   relative = FALSE       # for convergence function
) {
  ## Initialize
  converged <- FALSE
  i <- 0
  ## LOOP
  while (!converged & i <= maxIterations) {
    ## Update theta
    thetaNew <- theta - psiFn(theta)/psiPrimeFn(theta)
    ##
    ## Check convergence
    converged <- testConvergenceFn(thetaNew, theta,
                                   tolerance = tolerance,
                                   relative = relative)
    ## Update iteration
    theta <- thetaNew
    i <- i + 1
  }
  ## Return last value and whether converged or not
  list(theta = theta,
       converged = converged,
       iteration = i,
       fnValue = psiFn(theta)
       )
}
```

- Similar to our `gradientDescent` function we will need to write context-specific `psiFn(...)` and `psiPrimeFn(...)` functions if we want to use this in practice.
- We can reuse the `testConvergence` function that we wrote for use with `gradientDescent` in this context

```
testConvergence <- function(thetaNew, thetaOld, tolerance = 1e-10, relative = FALSE) {
    sum(abs(thetaNew - thetaOld)) < if (relative)
        tolerance * sum(abs(thetaOld)) else tolerance
}
```

Example (Weighted Average)

Let's consider the `facebook` dataset. Recall:

Variate	Value
`share`	the total (lifetime) number of times the post was shared
`like`	the total (lifetime) number of times the post "liked"
`comment`	the total (lifetime) number of comments attached to the post
`All.interactions`	the sum of `share`, `like`, and `comment`
`Impressions`	the total (lifetime) number of times the post has been displayed, whether the post is clicked or not. The same post may be seen by a Facebook user several times (e.g. via a page update in their News Feed once, whenever a friend shares it, etc.).

```
facebook <- read.csv("../../Data/facebook.csv", header = TRUE)
### Remove all rows with missing data
fb <- na.omit(facebook)
head(fb[, c(2, 3, 4, 1, 6)])
```

```
##   share like comment All.interactions Impressions
## 1    17   79       4              100        5091
## 2    29  130       5              164       19057
## 3    14   66       0               80        4373
## 4   147 1572      58             1777       87991
## 5    49  325      19              393       13594
## 6    33  152       1              186       20849
```

- Interest lies in determining θ, the typical number of `likes` a post receives
  - Let's account for the `Impressions` for each post since a post that is seen a lot will tend to have more likes
  - Let's do this by weighting `like` by the inverse of `Impressions` so that the more often a post is seen, the lower the weight we assign to its number of `likes`
- The solution to the weighted sum

$$\psi(\theta; \mathcal{P}) = \sum_{u \in \mathcal{P}} w_u (y_u - \theta) = 0$$

  can now be found via Newton's Method by defining the appropriate `psi(...)` and `psiPrime(...)` functions.

- In this case $\psi'(\theta; \mathcal{P})$ is given by

$$\psi'(\theta; \mathcal{P}) = -\sum_{u \in \mathcal{P}} w_u$$

- And the R implementations are as follows:

```
### Here we create both functions at once
createPsiFns <- function(y, wt) {
    psi <- function(theta = 0) {
        sum(wt * (y - theta))
    }
    psiPrime <- function(theta = 0) {
        -sum(wt)
    }
    list(psi = psi, psiPrime = psiPrime)
}

### Create them for the particular data
psiFns <- createPsiFns(y = fb$like, wt = 1/fb$Impressions)
psi <- psiFns$psi
psiPrime <- psiFns$psiPrime
```

- Passing these functions into `Newton(...)` yields the following:

```
result <- Newton(theta = mean(fb$like), psiFn = psi, psiPrimeFn = psiPrime)
print(result)
```

```
## $theta
## [1] 78.46845
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 2
##
## $fnValue
## [1] 7.962381e-16
```

- **Note:** we have a closed form expression for the weighted average

$$\frac{\sum_{u \in \mathcal{P}} w_u y_u}{\sum_{u \in \mathcal{P}} w_u}$$

and so the result above should agree with this closed form solution:

```
y <- fb$like
wt <- 1/fb$Impressions
c(NewtonSol = result$theta, weightedAve = sum(wt * y)/sum(wt))
```

```
##   NewtonSol weightedAve
##    78.46845    78.46845
```

- **Question:** Why did this converge in two iterations?

Summary

- We illustrated how some attributes can be defined implicitly through systems of equations.
- Then we considered how to use Newton's Method to numerically solve these equations.

Exercise

- Try out some other ψ functions for location estimation for the facebook example
  - e.g. Andrew's sine function, Tukey's bisquare, Huber's psi, or Hampel's psi

2.3.3b The Newton-Raphson Method

Overview

- When $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_k)'$ is a vector-valued attribute, the multivariate analog to Newton's Method is referred to as the **Newton-Raphson Method**.
- Here we examine the Newton-Raphson algorithm as a means to determining implicitly defined functions
  - We also discuss its connection with Gradient Descent.

When $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_k)'$ is a vector-valued attribute, the multivariate analog to Newton's Method is referred to as the **Newton-Raphson Method**. In this case we want to determine the vector which solves

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \mathbf{0}$$

Since $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})$ is a differentiable $k \times 1$ *vector* $(\psi_1, \psi_2, \ldots, \psi_k)'$ we let

$$\boldsymbol{\psi}'(\boldsymbol{\theta}; \mathcal{P}) = \frac{\partial\ \boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial \psi_1}{\partial \theta_1} & \frac{\partial \psi_1}{\partial \theta_2} & \cdots & \frac{\partial \psi_1}{\partial \theta_k} \\ \frac{\partial \psi_2}{\partial \theta_1} & \frac{\partial \psi_2}{\partial \theta_2} & \cdots & \frac{\partial \psi_2}{\partial \theta_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \psi_k}{\partial \theta_1} & \frac{\partial \psi_k}{\partial \theta_2} & \cdots & \frac{\partial \psi_k}{\partial \theta_k} \end{bmatrix}$$

be the $k \times k$ *matrix* of partial derivatives. This matrix is called the *Jacobian* matrix of $\boldsymbol{\psi}$

- Given a current guess $\widehat{\boldsymbol{\theta}}_i$ we can use a linear function to approximate the function $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})$
  - A first-order approximation of $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P})$ in the vicinity of $\widehat{\boldsymbol{\theta}}_i$ can be written as

$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) \approx \boldsymbol{\psi}(\widehat{\boldsymbol{\theta}}_i; \mathcal{P}) + \boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i; \mathcal{P}) \times \left(\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}}_i\right)$$

- Then the vector at which the linear approximation is equal to zero is

$$\boldsymbol{\theta} \approx \widehat{\boldsymbol{\theta}}_i - \left[\boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})\right]^{-1}\ \boldsymbol{\psi}(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})$$

which suggests the iterative **Newton-Raphson** algorithm for root finding.

The Newton-Raphson Algorithm

Given an initial value $\widehat{\boldsymbol{\theta}}_0$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:

   a. **Update** the iterate:
   $$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}_i - \left[\boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})\right]^{-1} \boldsymbol{\psi}(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})$$

   b. **Converged?**
   if the iterates are not changing, then **return**
   else $i \leftarrow i + 1$ and repeat LOOP.
3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

R Code for the Newton-Raphson Algorithm

```
NewtonRaphson <- function(theta, psiFn, psiPrimeFn, dim, testConvergenceFn = testConvergence,
    maxIterations = 100, tolerance = 1e-06, relative = FALSE) {
    if (missing(theta)) {
        ## need to figure out the dimensionality
        if (missing(dim)) {
            dim <- length(psiFn())
        }
        theta <- rep(0, dim)
    }
    converged <- FALSE
    i <- 0
    while (!converged & i <= maxIterations) {
        thetaNew <- theta - solve(psiPrimeFn(theta), psiFn(theta))
        converged <- testConvergenceFn(thetaNew, theta, tolerance = tolerance,
            relative = relative)

        theta <- thetaNew
        i <- i + 1
    }
    ## Return last value and whether converged or not
    list(theta = theta, converged = converged, iteration = i, fnValue = psiFn(theta))
}
```

Note that `NewtonRaphson(...)` is identical to `Newton(...)` with two important exceptions:

1. First, since this is Newton's method in arbitrary dimensions, we must be given the dimensionality either implicitly from `theta` or explicitly via `dim`.

    - If neither of these arguments are given, it is inferred by evaluating the `psiFn(...)` once with no arguments.

2. The update for `theta` is attained by using `solve(...)` instead.

    - This is needed because

$$\left[\boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i;\mathcal{P})\right]^{-1}$$

is the value of $\mathbf{X}$ that solves the system of equations

$$\left[\boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i;\mathcal{P})\right]\ \mathbf{X} = \boldsymbol{\psi}(\widehat{\boldsymbol{\theta}}_i;\mathcal{P})$$

which is precisely what the function `solve(...)` does.

Example (Rosenbrock Function)

Suppose we are interested in finding $\boldsymbol{\theta} = (\theta_1, \theta_2)$ which minimizes the Rosenbrock function

$$\rho\left(\boldsymbol{\theta}\right) = \left(1-\theta_1\right)^2 + 100\left(\theta_2 - \theta_1^2\right)^2$$

With the problem framed as *solving a system of equations*, the roots of the gradient are of importance:

$$\boldsymbol{\psi}\left(\boldsymbol{\theta}\right) = \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \equiv \boldsymbol{g} = \nabla\rho(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial\rho}{\partial\theta_1} \\ \frac{\partial\rho}{\partial\theta_2} \end{bmatrix} = \begin{bmatrix} 400\theta_1^3 - 400\theta_1\theta_2 + 2\theta_1 - 2 \\ \\ -200\theta_1^2 + 200\theta_2 \end{bmatrix}$$

And for Newton-Raphson we need the matrix of partial derivatives of $\boldsymbol{\psi}\left(\boldsymbol{\theta}\right)$ (i.e., the Jacobian of $\boldsymbol{\psi}\left(\boldsymbol{\theta}\right)$):

$$\boldsymbol{\psi}'(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial\psi_1}{\partial\theta_1} & \frac{\partial\psi_1}{\partial\theta_2} \\ \frac{\partial\psi_2}{\partial\theta_1} & \frac{\partial\psi_2}{\partial\theta_2} \end{bmatrix} = \begin{bmatrix} 1200\ \theta_1^2 - 400\ \theta_2 + 2 & -400\ \theta_1 \\ -400\ \theta_1 & 200 \end{bmatrix}$$

```r
psiPrime <- function(theta = c(0, 0)) {
    val = matrix(0, nrow = length(theta), ncol = length(theta))
    val[1, 1] = 1200 * theta[1]^2 - 400 * theta[2] + 2
    val[1, 2] = -400 * theta[1]
    val[2, 1] = -400 * theta[1]
    val[2, 2] = 200
    return(val)
}
```

- Recall the `R` functions for the objective function and gradient

```
rho <- function(theta) {
    (1 - theta[1])^2 + 100 * (theta[2] - theta[1]^2)^2
}
g <- function(theta) {
    c(400 * theta[1]^3 - 400 * theta[1] * theta[2] + 2 * theta[1] - 2, -200 *
        theta[1]^2 + 200 * theta[2])
}
```

The value of $\widehat{\theta}$ as determined by Newton-Raphson is as follows:

```
NRresult <- NewtonRaphson(theta = c(0, 0), psiFn = g, psiPrimeFn = psiPrime)

print(NRresult)
```

```
## $theta
## [1] 1 1
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 3
##
## $fnValue
## [1] 0 0
```

Let's compare that to the value of $\widehat{\theta}$ we found with gradient descent:

```
GDresult = gradientDescent(theta = c(0, 0), rhoFn = rho, gradientFn = g, lineSearchFn = gridLineSearch,
    lambdaStepsize = 1e-04, testConvergenceFn = testConvergence, maxIterations = 1000)

print(GDresult)
```

```
## $theta
## [1] 0.9940877 0.9882748
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 254
##
## $fnValue
## [1] 3.537011e-05
```

- Let's check the sensitivity of these results to the starting value $\boldsymbol{\theta}_0$
  - Ideally an algorithm's performance shouldn't depend too much on the initial guess
  - But in practice, an accurate initial guess can make a huge difference

- Rather than starting at $\boldsymbol{\theta}_0 = (0, 0)$, let's start the algorithms at $\boldsymbol{\theta}_0 = (-2, -1)$

Newton-Raphson:

```r
NRresult <- NewtonRaphson(theta = c(-2, -1), psiFn = g, psiPrimeFn = psiPrime,
    maxIterations = 10^4)

print(NRresult)
```

```
## $theta
## [1] 1 1
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 6
##
## $fnValue
## [1] 0 0
```

Gradient Descent:

```r
GDresult = gradientDescent(theta = c(-2, -1), rhoFn = rho, gradientFn = g, lineSearchFn = gridLineSearc
    lambdaStepsize = 1e-04, testConvergenceFn = testConvergence, maxIterations = 10000)

print(GDresult)
```

```
## $theta
## [1] 1.000379 1.000647
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 6657
##
## $fnValue
## [1] 1.364339e-06
```

Connection between Newton-Raphson and Gradient Descent

- Notice that when the objective function of interest is $\rho(\boldsymbol{\theta}; \mathcal{P})$ then $\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \nabla\rho(\boldsymbol{\theta}; \mathcal{P}) = \boldsymbol{g}$ and the system of equations
$$\boldsymbol{\psi}(\boldsymbol{\theta}; \mathcal{P}) = \mathbf{0}$$
is equivalent to
$$\nabla\rho(\boldsymbol{\theta}; \mathcal{P}) = \mathbf{0}$$

- Also notice that the updating equation associated with Newton-Raphson is given by
$$\widehat{\boldsymbol{\theta}}_{i+1} = \widehat{\boldsymbol{\theta}}_i - \left[\boldsymbol{\psi}'(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})\right]^{-1} \ \boldsymbol{\psi}(\widehat{\boldsymbol{\theta}}_i; \mathcal{P})$$
which, in light of the previous point, can be rewritten as
$$\widehat{\boldsymbol{\theta}}_{i+1} = \widehat{\boldsymbol{\theta}}_i - H_i^{-1} \ \boldsymbol{g}_i$$
where
$$H_i = \left.\begin{bmatrix} \frac{\partial^2\rho}{\partial\theta_1^2} & \frac{\partial^2\rho}{\partial\theta_1\theta_2} & \cdots & \frac{\partial^2\rho}{\partial\theta_1\theta_k} \\ \frac{\partial^2\rho}{\partial\theta_2\theta_1} & \frac{\partial^2\rho}{\partial\theta_2^2} & \cdots & \frac{\partial^2\rho}{\partial\theta_2\theta_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2\rho}{\partial\theta_k\theta_1} & \frac{\partial^2\rho}{\partial\theta_k\theta_2} & \cdots & \frac{\partial^2\rho}{\partial\theta_k^2} \end{bmatrix}\right|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_i}$$
is the *Hessian* of ρ (i.e., the matrix of second partial derivatives of ρ with respect to $\theta_1, \theta_2, \ldots, \theta_k$)

- Recall that the updating equation associated with gradient descent is given by
$$\begin{aligned} \widehat{\boldsymbol{\theta}}_{i+1} &= \widehat{\boldsymbol{\theta}}_i - \widehat{\lambda}_i \ \boldsymbol{d}_i \\ &= \widehat{\boldsymbol{\theta}}_i - \frac{\widehat{\lambda}_i}{\|\boldsymbol{g}_i\|}\boldsymbol{g}_i \end{aligned}$$

- Thus Newton-Raphson applied to the gradient of an objective function is essentially equivalent to gradient descent applied directly to that objective function but with step sizes modulated by the Hessian of the objective function

Summary

- We reviewed the Newton-Raphson algortihm and how to implement it in R.
- We discussed the connection between Newton-Raphson and Gradient Descent.

Exercise

- From the Implicit Attribute exercises try problems 3(g), 10(j) and 11(e).

2.3.4 Iteratively Reweighted Least Squares

Overview

- Iteratively reweighted least squares (IRLS) provides an iterative alogrithm for finding $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$, the solution to
$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}\in\boldsymbol{\Theta}}{\operatorname{argmin}} \sum_{u\in\mathcal{P}} \rho(y_u - \alpha - \beta(x_u - c))$$
- We consider how to implement this algorithm in R and consider two examples using least squares and robust regression.

When $\boldsymbol{\theta} = (\alpha, \beta)'$ is a vector valued attribute associated with a linear regression, **iteratively reweighted least squares (IRLS)** provides a conventient iterative method for finding $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$, the solution to

$$\underset{\boldsymbol{\theta}\in\boldsymbol{\Theta}}{\operatorname{argmin}} \sum_{u\in\mathcal{P}} \rho(y_u - \alpha - \beta(x_u - c))$$

Differentiating this with respect to α and β and setting the result equal to zero yields

$$\sum_{u\in\mathcal{P}} \rho'(y_u - \alpha - \beta(x_u - c)) \begin{pmatrix} 1 \\ x_u - c \end{pmatrix} = \mathbf{0}$$

and $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$ is found by solving this system of equations.

If we let $r_u = y_u - \alpha - \beta(x_u - c)$ and $\boldsymbol{z}_u = (1,\ x_u - c)'$, then the system above becomes

$$\sum_{u\in\mathcal{P}} \rho'(r_u)\ \boldsymbol{z}_u = \mathbf{0}$$

- Note that in WLS $\rho(r_u) = w_u r_u^2$ and $\rho'(r_u) = 2w_u r_u$ and so the system reduces to

$$\sum_{u\in\mathcal{P}} w_u r_u\ \boldsymbol{z}_u = \mathbf{0}$$

- With the IRLS algorithm we will exploit the fact that the general objective function minimization problem can be recast as a weighted least squares problem
- The general equation can be made to look like the WLS equation as follows:

$$
\begin{aligned}
\mathbf{0} &= \textstyle\sum_{u \in \mathcal{P}} \rho'(r_u)\ \boldsymbol{z}_u \\
&= \textstyle\sum_{u \in \mathcal{P}} \left(\frac{\rho'(r_u)}{r_u}\right)\ r_u\ \boldsymbol{z}_u \\
&= \textstyle\sum_{u \in \mathcal{P}} w_u r_u\ \boldsymbol{z}_u
\end{aligned}
$$

where $w_u = \frac{\rho'(r_u)}{r_u}$ is the weight for unit u provided $r_u \neq 0$

- Why did we do this?

- If we had some initial value for the residuals r_u (and hence the weights w_u) we could solve this equation in closed form yielding a value for $\boldsymbol{\theta}$, call it $\widehat{\boldsymbol{\theta}}_1 = (\widehat{\alpha}_1, \widehat{\beta}_1)'$
  - This requires initial values for α and β
  - Since $\boldsymbol{\theta}_0 = (\alpha_0, \beta_0)$ is likely far from $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$ we should iterate and update our values of the residuals (and hence weights) with $\widehat{\boldsymbol{\theta}}_1 = (\widehat{\alpha}_1, \widehat{\beta}_1)'$
  - This process can be repeated until the estimates of α and β converge
  - This process is called **iteratively reweighted least squares**
- We can generalize this algorithm by expressing the problem in terms of the vector-valued attribute $\boldsymbol{\theta} = (\alpha,\ \beta)'$: $r_u = y_u - \boldsymbol{z}_u'\boldsymbol{\theta}$.

The Algorithm

Given an initial value $\widehat{\boldsymbol{\theta}}_0$

1. **Initialize** ; $i \leftarrow 0$;
2. LOOP:

   a. **Construct residuals and weights** for all $u \in \mathcal{P}$

   $$r_u = y_u - \boldsymbol{z}_u'\boldsymbol{\theta}_i \qquad \text{and} \qquad w_u = \frac{\rho'(r_u)}{r_u}$$

   b. **Solve** the weighted least squares problem, i.e., find $\widehat{\boldsymbol{\theta}}$, the value of $\boldsymbol{\theta}$ such that:

   $$\sum_{u \in \mathcal{P}} w_u(y_u - \boldsymbol{z}_u'\boldsymbol{\theta})\ \boldsymbol{z}_u = \mathbf{0}$$

c. **Update** the parameter:

$$\widehat{\boldsymbol{\theta}}_{i+1} \leftarrow \widehat{\boldsymbol{\theta}}$$

d. **Converged?**
if the iterates are not changing, then **return**
else $i \leftarrow i+1$ and repeat LOOP.

3. **Return**: $\widehat{\boldsymbol{\theta}} = \widehat{\boldsymbol{\theta}}_i$

- The algorithm above works for any vector-valued attribute $\boldsymbol{\theta}$ that arises in the context of a **linear response model**:

$$y_u = \mathbf{z}_u' \boldsymbol{\theta} + r_u$$

R code for IRLS

```
irls <- function(y, x, theta, rhoPrimeFn,
                 dim = 2, delta = 1E-10,
                 testConvergenceFn = testConvergence,
                 maxIterations = 100,   # maximum number of iterations
                 tolerance = 1E-6,      # parameters for the test
                 relative = FALSE       # for convergence function
){if (missing(theta)) {theta <- rep(0, dim)}
  ## Initialize
  converged <- FALSE
  i <- 0
  N <- length(y)
  wt <- rep(1,N)
  ## LOOP
  while (!converged & i <= maxIterations) {
    ## get residuals
    resids <- getResids(y, x, wt, theta)
    ## update weights  (should check for zero resids)
    wt <- getWeights(resids, rhoPrimeFn, delta)
    ## solve the least squares problem
    thetaNew <- getTheta(y, x, wt)
    ##
    ## Check convergence
    converged <- testConvergenceFn(thetaNew, theta,
                                   tolerance = tolerance,
                                   relative = relative)
    ## Update iteration
    theta <- thetaNew
    i <- i + 1
  }
  ## Return last value and whether converged or not
  list(theta = theta, converged = converged, iteration = i)
}
```

Example 1 (Least Squares)

- Let us perform **least squares** regression to fit the following simple linear regression model

$$y_u = \alpha + \beta(x_u - \overline{x}) + r_u$$

  but by using IRLS as the method of estimation.

- In order to use the `irls` function we must first write the functions `getWeights(...)` and `getTheta(...)`, which for any problem are:

```
getWeights <- function(resids, rhoPrimeFn, delta = 1e-12) {
    ## for calculating weights, minimum |residual| will be delta
    smallResids <- abs(resids) <= delta
    ## take care to preserve sign (in case rhoPrime not symmetric)
    resids[smallResids] <- delta * sign(resids[smallResids])
    ## calculate and return weights
    rhoPrimeFn(resids)/resids
}

getTheta <- function(y, x, wt) {
    theta <- numeric(length = 2)
    ybarw <- sum(wt * y)/sum(wt)
    xbarw <- sum(wt * x)/sum(wt)
    theta[1] <- ybarw - (sum(wt * (x - xbarw) * y)/sum(wt * (x - xbarw)^2)) *
        (xbarw - mean(x))
    theta[2] <- sum(wt * (x - xbarw) * y)/sum(wt * (x - xbarw)^2)
    ## return theta
    theta
}
```

We must also define `getResids(...)` and a `rhoPrime` function, which for ordinary least squares regression are both based on

$$r_u = y_u - \alpha - \beta(x - \overline{x}), \qquad \rho(r_u) = r_u^2, \qquad \rho'(r_u) = 2r_u$$

```
getResids <- function(y, x, wt, theta) {
    xbar <- mean(x)
    alpha <- theta[1]
    beta <- theta[2]
    ## resids are
    y - alpha - beta * (x - xbar)
}

LSrhoPrime <- function(resid) {
    2 * resid
}
```

Let's now calculate $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$ for the `Where's Waldo` data using IRLS:

```
waldo <- read.csv("../../Data/wheres-waldo-locations.csv", header = TRUE)
IRLSresult <- irls(waldo$Y, waldo$X, theta = c(0, 0), rhoPrimeFn = LSrhoPrime)

print(IRLSresult)
```

```
## $theta
## [1] 3.8753064 0.1429041
##
## $converged
## [1] TRUE
##
## $iteration
## [1] 2
```

Let's also compare this result to that which is obtained by using `lm` in R:

```
lm(waldo$Y ~ I(waldo$X - mean(waldo$X)))$coef
```

```
##                (Intercept) I(waldo$X - mean(waldo$X))
##                  3.8753064                  0.1429041
```

Example 2 (Robust Regression)

- Recall the Huber Function was defined as

$$\rho_k(r) = \begin{cases} \frac{1}{2}r^2 & \text{for } |r| \leq k, \\ k\,|r| - \frac{1}{2}k^2, & \text{otherwise.} \end{cases}$$

  and implemented in `R` with the function `huber.fn`

- Recall also that the Huber Function's derivative is given by

$$\rho_k'(r) = \begin{cases} r & \text{for } |r| \leq k, \\ \text{sign}(r) \times k, & \text{otherwise.} \end{cases}$$

  and implemented in `R` with the function `huber.fn.prime`:

```
huber.fn.prime <- function(resid, k = 1.345) {
    val = resid
    subr = abs(resid) > k
    val[subr] = k * sign(resid[subr])
    return(val)
}
```

- Let's perform **robust regression** on the `Animals` data but by using IRLS as our method of estimation

```
data(Animals2, package = "robustbase")
result <- irls(log(Animals2$brain), log(Animals2$body), theta = c(1, 1), rhoPrimeFn = huber.fn.prime,
    maxIterations = 100)
print(result)
```

```
## $theta
## [1] 3.3325141 0.6885898
##
```

```
## $converged
## [1] TRUE
##
## $iteration
## [1] 8
```

- Compare this result to that which is obtained by `rlm` function from the `MASS` package in `R`.

```r
library(MASS)
temp = rlm(log(Animals2$brain) ~ I(log(Animals2$body) - mean(log(Animals2$body))),
    psi = "psi.huber")
temp$coef
```

```
##                                      (Intercept)
##                                        3.3405534
## I(log(Animals2$body) - mean(log(Animals2$body)))
##                                        0.6985483
```

Summary

- Iteratively reweighted least squares (IRLS) provides an iterative algorithm for finding $\widehat{\boldsymbol{\theta}} = (\widehat{\alpha}, \widehat{\beta})'$, the solution to

$$\underset{\boldsymbol{\theta}\in\boldsymbol{\Theta}}{\operatorname{argmin}} \sum_{u\in\mathcal{P}} \rho(y_u - \alpha - \beta(x_u - c))$$

- We considered two examples of IRLS using least squares and robust regression.

Exercises

From the Implicit Attribute exercises try problems 6 and 7.